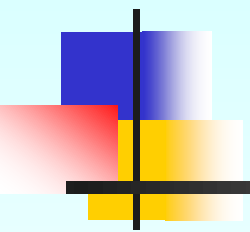
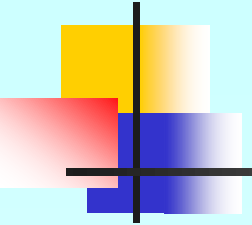


Lecture 5



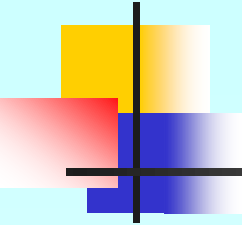
Spatial Analysis with Raster Data

第 5 讲 栅格数据的空间分析



Main Contents

- 1. Concepts**
- 2. Some essential problems in SA with Raster**
- 3. Function Framework**
- 4. Integrated Analysis: Map Algebra**
- 5. Applications**



1. Concepts

1.1 Understanding Raster Data

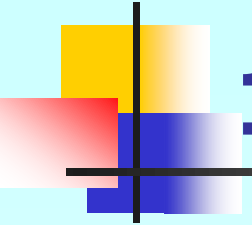
1.2 Raster Bands

1.3 Resolution: Cell Size

1.4 View of Time and Space

1.5 Mathematic Foundation of Raster Data

1.6 Computer Representation of Raster Data



1.1 Understanding Raster Data

□ 栅格数据一般被分为两类

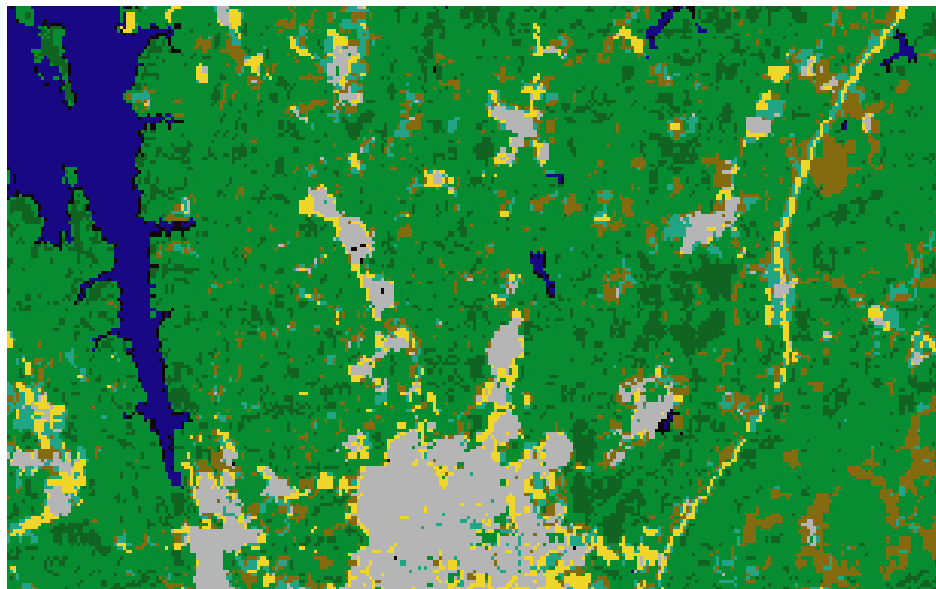
■ 专题栅格数据（**thematic data**）：值表示某种测量值或分类，如高程、污染浓度、人口或土地覆被类型

■ 图像数据（**image data**）：图像中单元的值代表了诸如卫星图像或扫描照片的反射或发射的光或能量

□ 空间分析工具主要是针对专题栅格数据的。

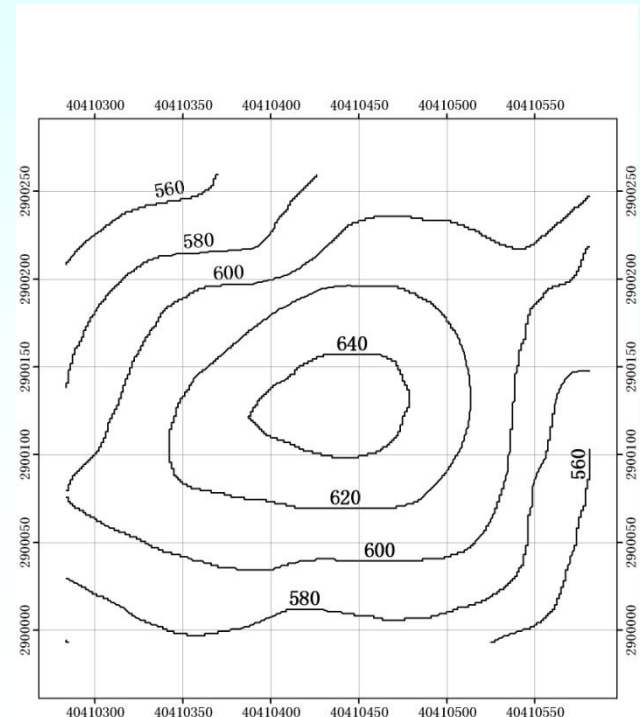
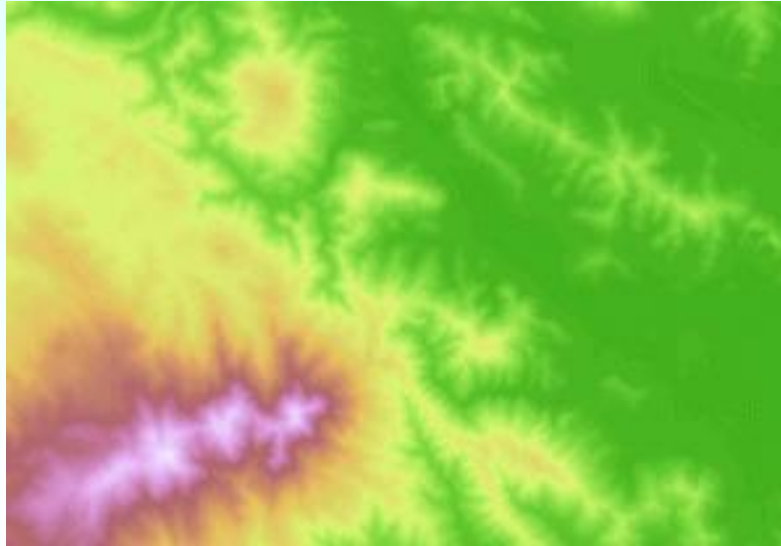
1.1 Understanding Raster Data

Rasters representing thematic data can be derived from analyzing other data. A common analysis application is classifying a satellite image by land-cover categories.



1.1 Understanding Raster Data

Rasters are well suited for representing data that **changes continuously** across a landscape (surface).



1.1 Understanding Raster Data

- Rasters used as attributes of a feature may be digital photographs, scanned documents, or scanned drawings related to a geographic object or location.
- 缺点: **flat** (扁平化), 如路灯与榕树的叠合, 只保留了其中一种地物的像素值。



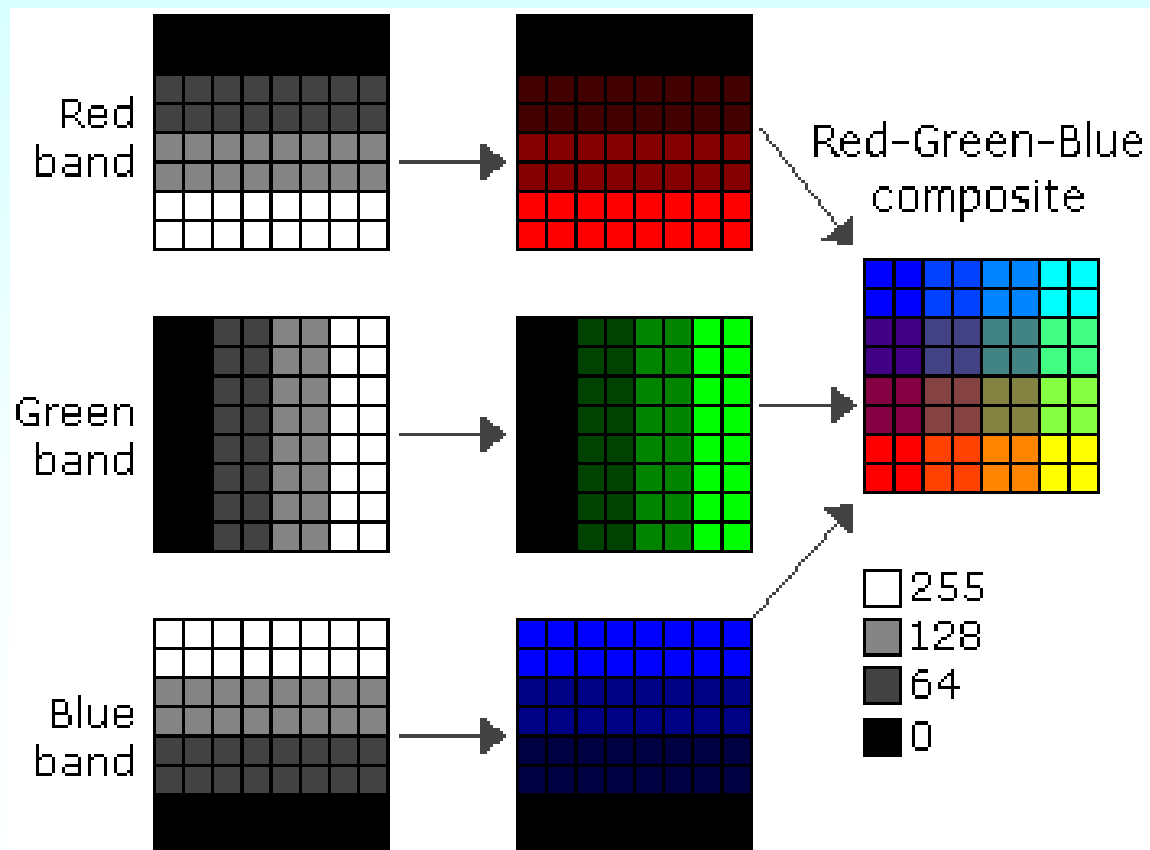
1.1 Understanding Raster Data

Remotely sensed image used as a base map.



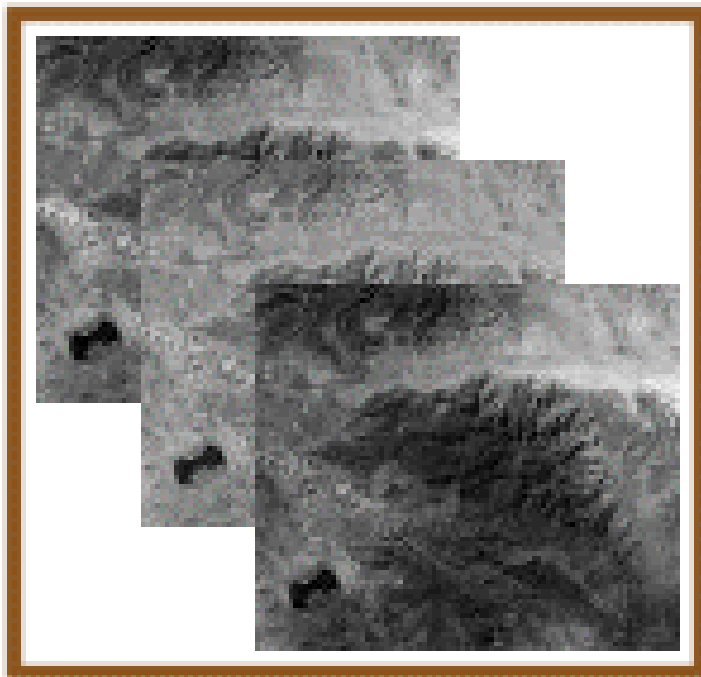
1.2 Raster bands

RGB Composition



1.2 Raster bands

Multiband
raster dataset

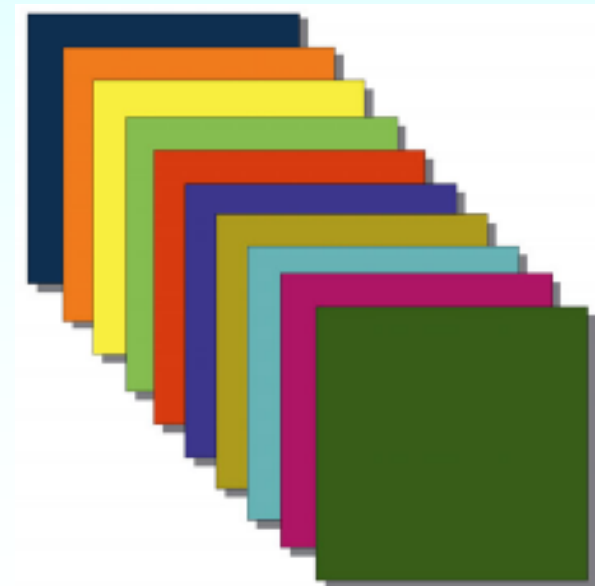
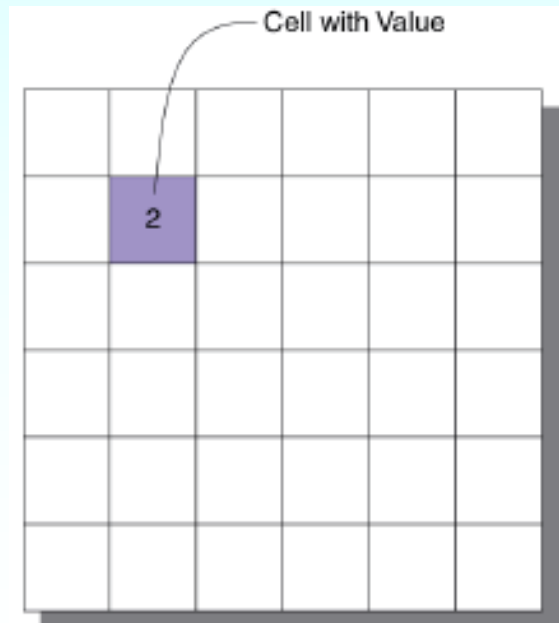
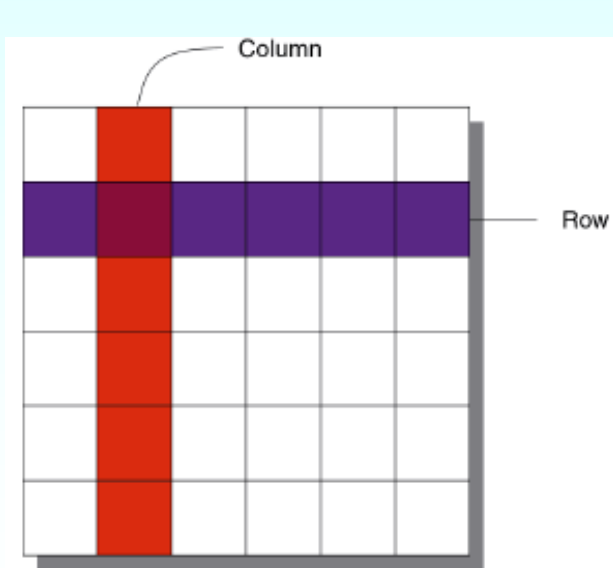


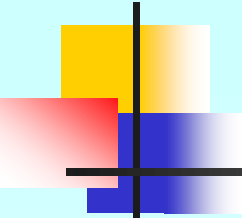
RGB composite



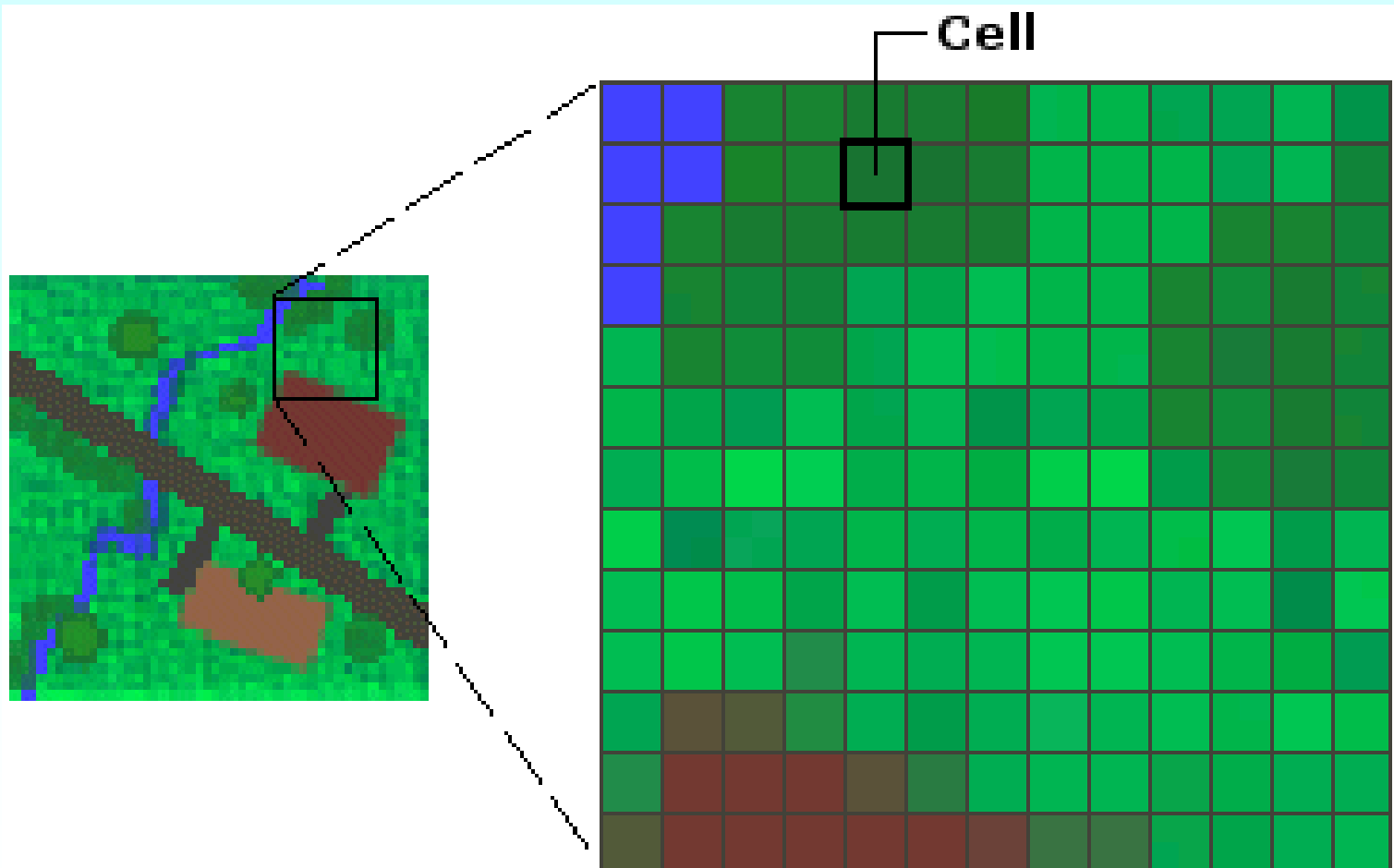
1.2 Raster bands

所有空间分析模块的函数，处理的都是栅格数据集的第一个波段。

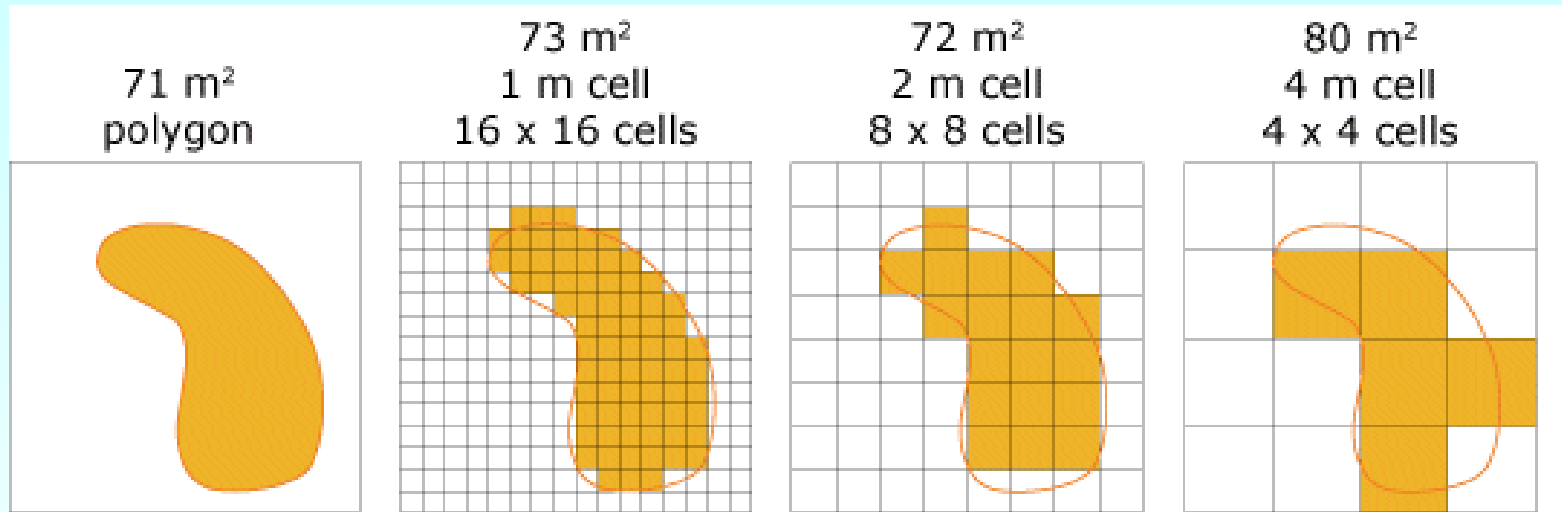




1.3 Resolution—cell size



1.3 Resolution—cell size



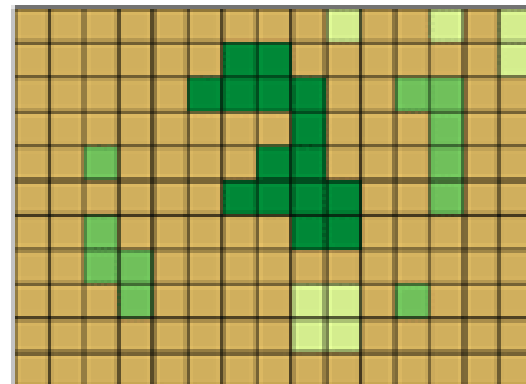
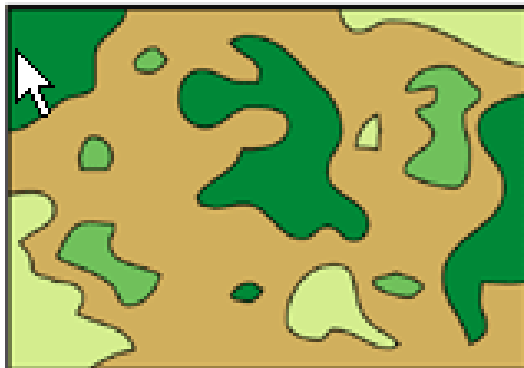
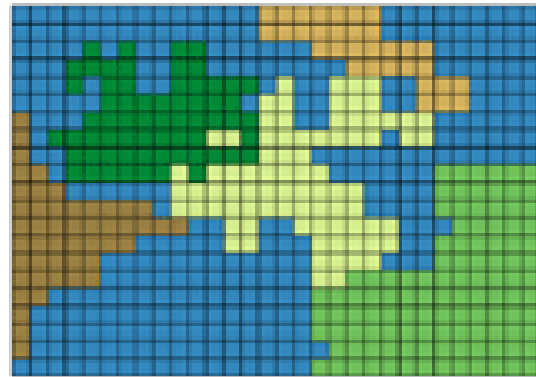
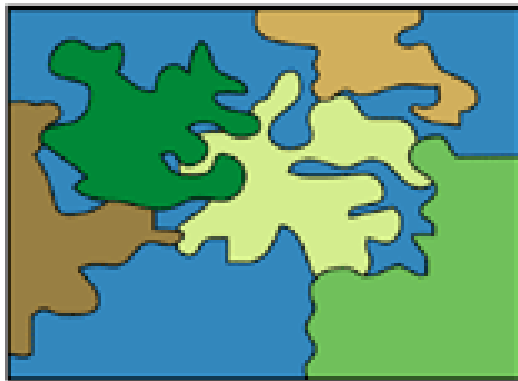
多来源栅格数据输入到建模软件（如水文模型）中计算时，栅格数据的套合误差问题：可能导致模型计算出错。

- Smaller cell size
- Higher resolution
- Higher feature spatial accuracy
- Slower display
- Slower processing
- Larger file size

- Larger cell size
- Lower resolution
- Lower feature spatial accuracy
- Faster display
- Faster processing
- Smaller file size

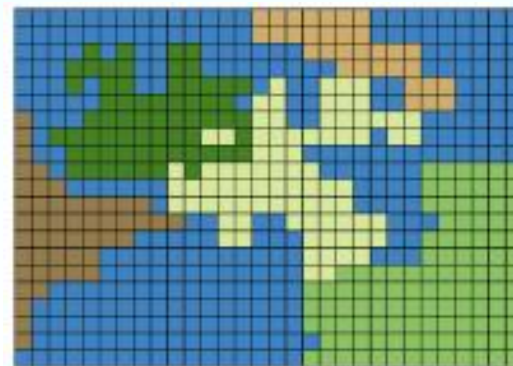
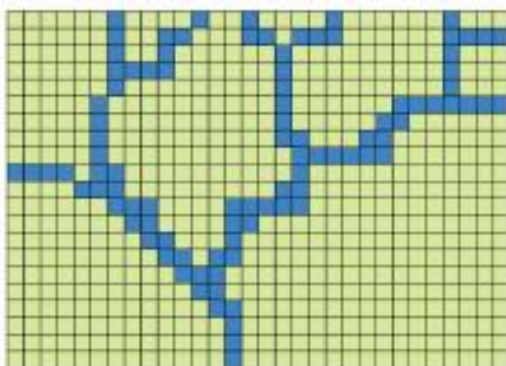
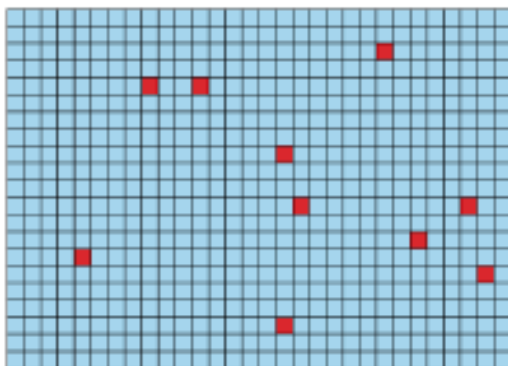
1.3 Resolution—cell size

找找： Information loss during the transformation between the vector and raster data.



1.3 Resolution—cell size

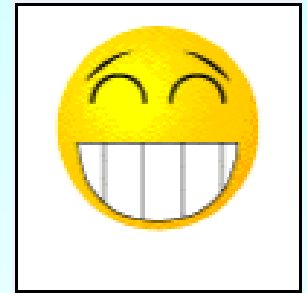
- 空间位置的精度：像元代表地物的位置
 - 思考题：遥感影像配准精度达**0.5**个像元是什么意思？
- 属性值的精度：整型、浮点型
- 长度、面积的精度：
 - 思考题？具有面积的点对象的面积精度是多少？



1.4 View of the Time and Space

■ Entity View: Container of entities , the index of entities.

- Points, Polylines, Polygons and Bodies
- With vague or explicit boundary
- In continuous, abstracted form
- Such as: Pond, Lake, Reservoir, Lagoon

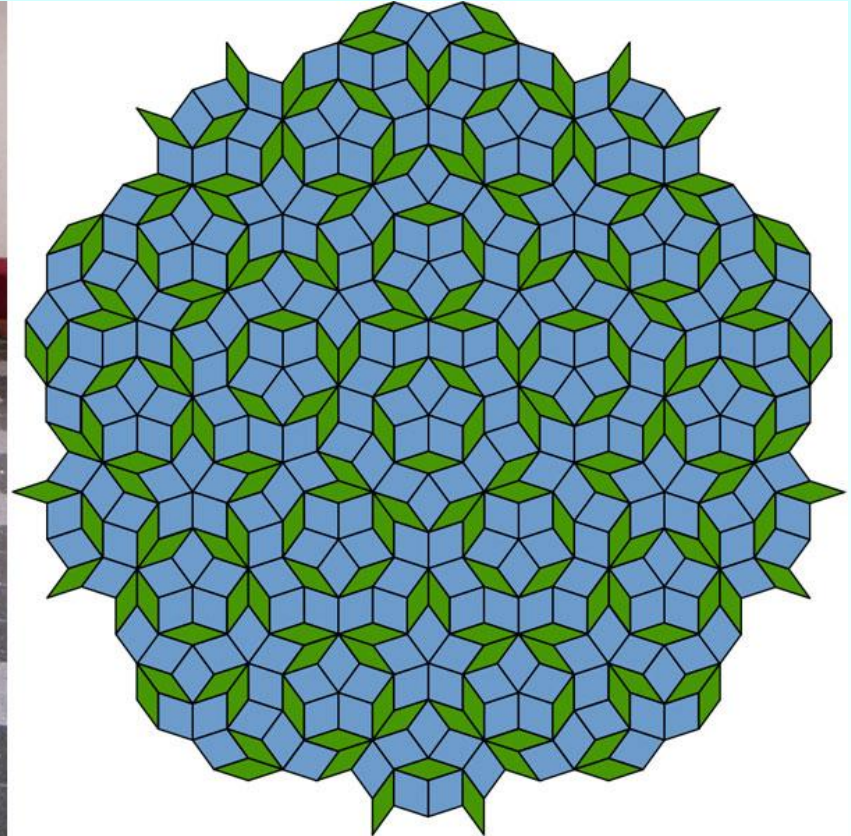


■ Field View: A field in which many kind of phenomena interacting or mixed

- Just Raster, image
- The distribution of one or several kind of phenomenon
- Continuous or discrete
- Such as: DEM, Remotely sensed image, SST (Sea Surface Temperature)



1.5 Mathematic foundation of Raster



- 栅格，一定要正方形吗？
- Roger Penrose 彭罗斯铺砖法（Penrose Tiling）

1.5 Mathematic foundation of Raster

Roger Penrose

罗杰·彭罗斯 (Roger Penrose) 简介

1. 基本信息

- **出生:** 1931 年 8 月 8 日, 英国埃塞克斯郡科尔切斯特
- **国籍:** 英国
- **职业:** 数学物理学家、哲学家、诺贝尔奖得主
- **现职:** 牛津大学数学荣誉教授, 英国皇家学会会员

2. 教育背景

- **1952 年:** 伦敦大学学院 (UCL) 数学学士 (一等荣誉学位)
- **1957 年:** 剑桥大学圣约翰学院博士, 研究方向为代数几何, 博士论文《张量方法在代数几何中的应用》

3. 工作经历

- **1964–1973 年:** 伦敦大学伯克贝克学院应用数学教授
- **1973–1998 年:** 牛津大学鲁斯 - 波尔数学教授
- **1998 年至今:** 牛津大学数学荣誉教授
- **其他职位:** 国际广义相对论和引力学会主席 (1992–1995), 美国哲学学会会员 (2011 年起)

4. 主要贡献

- **黑洞与奇点理论:**
 - 1965 年提出**彭罗斯奇点定理**, 证明黑洞核心存在时空奇点 (物理定律失效点), 并与霍金共同发展宇宙奇点理论。
 - 提出**彭罗斯过程**, 解释黑洞通过旋转释放能量的机制。
 - 发明**彭罗斯图**, 直观展示黑洞附近时空结构。
- **扭量理论:**
 - 创立扭量理论 (Twistor Theory), 试图通过复几何统一量子力学与广义相对论。
- **彭罗斯镶嵌:**
 - 1974 年发现**彭罗斯铺砖法**, 用两种非周期性图案铺满平面, 启发准晶体研究。
- **意识与人工智能:**
 - 提出**Orch-OR 理论** (客观还原理论), 认为意识源于神经元微管中的量子引力效应。
 - 反对强人工智能, 认为人类意识超越算法, 代表作《皇帝的新脑》引发广泛争议。
- **宇宙学:**
 - 提出**共形循环宇宙学**, 认为宇宙经历无限循环的“大爆炸 — 膨胀 — 收缩 — 新大爆炸”周期。

5. 代表作品

- 《皇帝的新脑》(1989): 探讨人工智能与意识的物理基础, 质疑算法可完全模拟人类思维。
- 《心灵的阴影》(1994): 深化意识的量子机制理论。
- 《通向实在之路》(2004): 系统梳理数学与物理知识, 被誉为“现代科学百科全书”。
- 《时间的轮回》(2010): 阐述共形循环宇宙学。

6. 荣誉与奖项

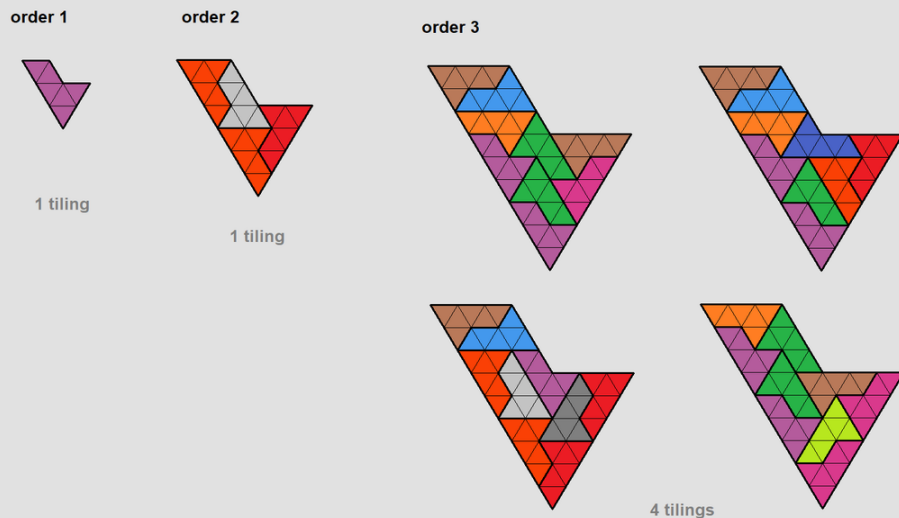
- **2020 年诺贝尔物理学奖:** 因“证明黑洞形成是广义相对论的可靠预测”获奖。
- **1988 年沃尔夫奖:** 与霍金共享, 表彰奇点理论贡献。
- **2008 年科普利奖章:** 英国皇家学会最高荣誉。
- **其他奖项:** 艾丁顿奖章 (1975)、狄拉克奖章 (1989)、爱因斯坦奖章 (1990) 等。

7. 家庭与个人

- **家庭背景:** 父亲为遗传学家莱昂内尔·彭罗斯, 叔叔为超现实主义艺术家罗兰·彭罗斯。
- **兴趣跨界:** 早年与父亲设计“彭罗斯三角形”等不可能图形, 与荷兰艺术家埃舍尔合作创作。

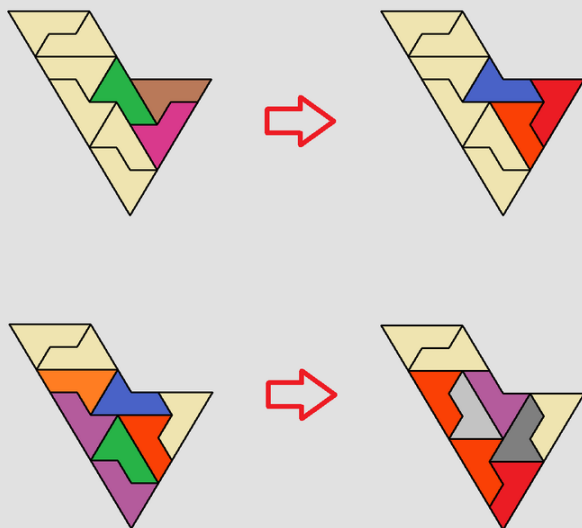
Number of tilings for sphinx order 1,2,3

三阶斯芬克斯铺砖的铺砌数目

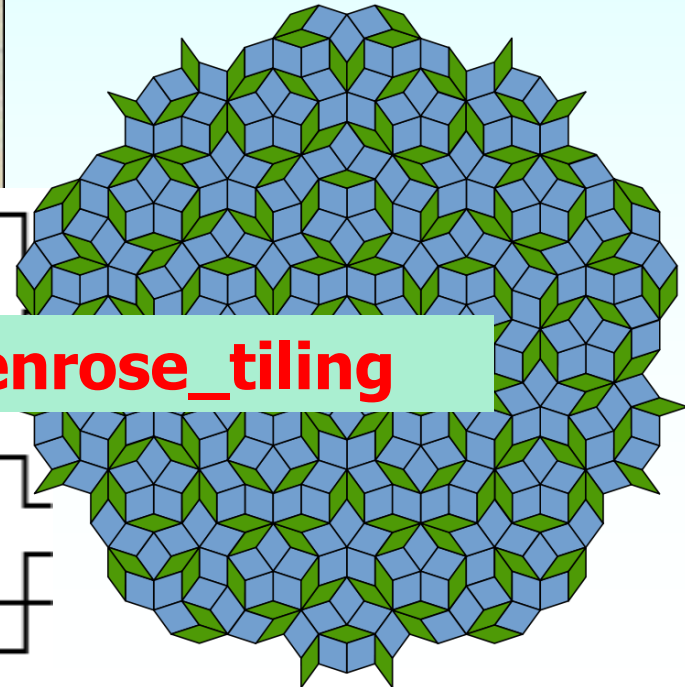
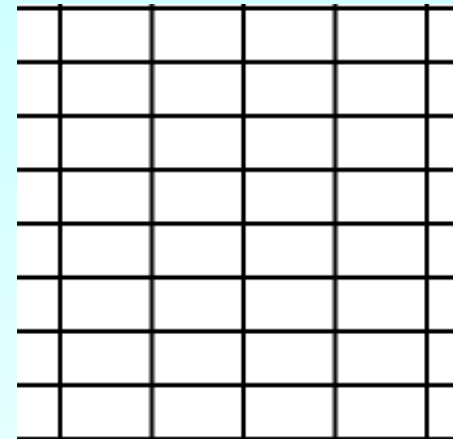


Internal symmetric transitional shapes for order 3

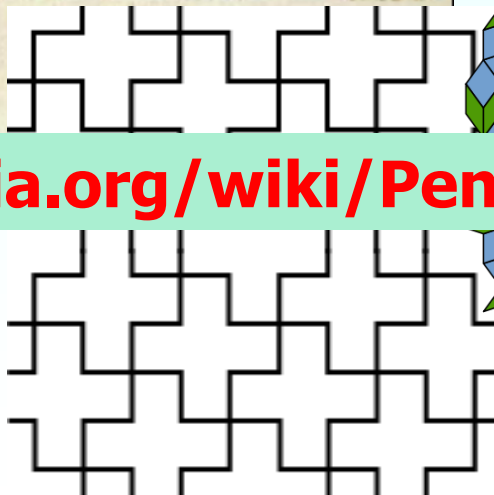
三阶内部对称过渡形状

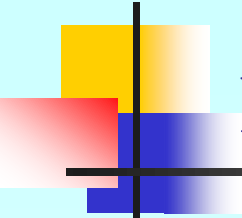


1.5 Mathematic foundation of Raster



https://en.wikipedia.org/wiki/Penrose_tiling





1.5 Mathematic foundation of Raster

■ **Spatial Tessellation:** 把 d 维欧氏空间 R^d 分割为覆盖 R^d 而没有缝隙与重叠的一系列 d 维区域。

■ **$D=3$:** 三维空间划分;

■ **$D=2$:** 平面划分;

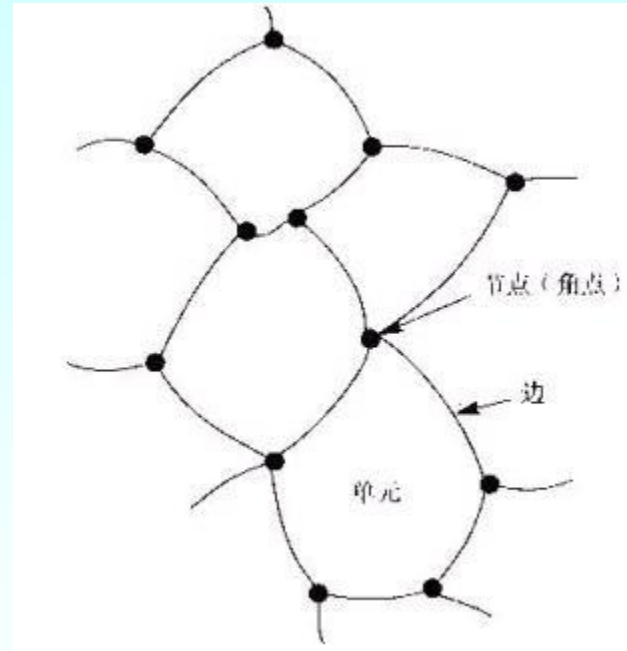
1.5 Mathematic foundation of Raster

■ **D=2: 平面划分;**

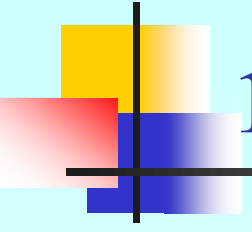
■ 划分单元 (2-d)

■ 边 (1-d)

■ 顶点 (0-d)



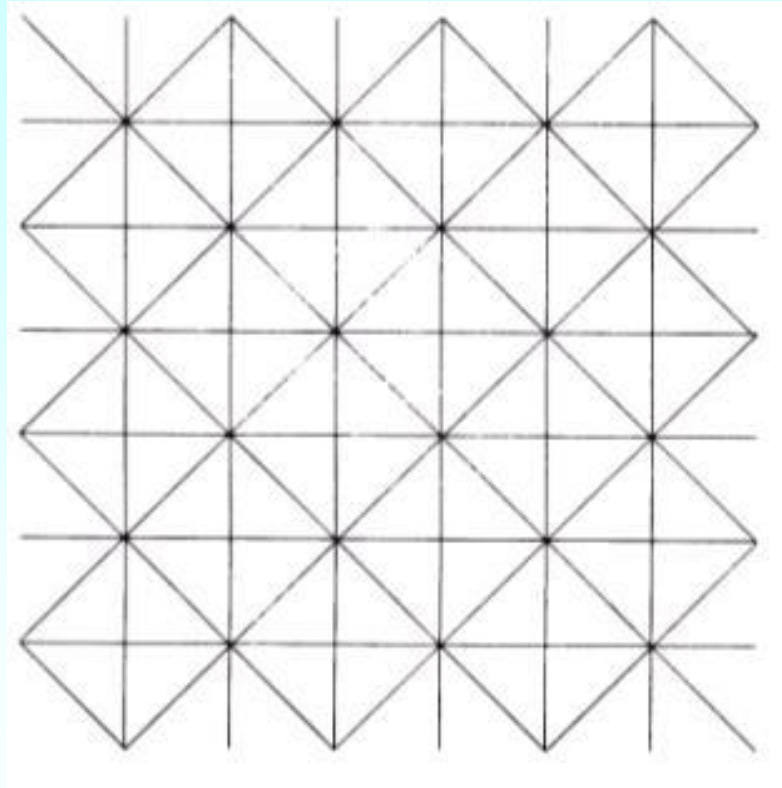
平面剖分的术语



1.5 Mathematic foundation of Raster

- 图论中的: Vertex, Edge, Graph, dimension;
- 矢量数据中的: Vertic, Node, Polyline

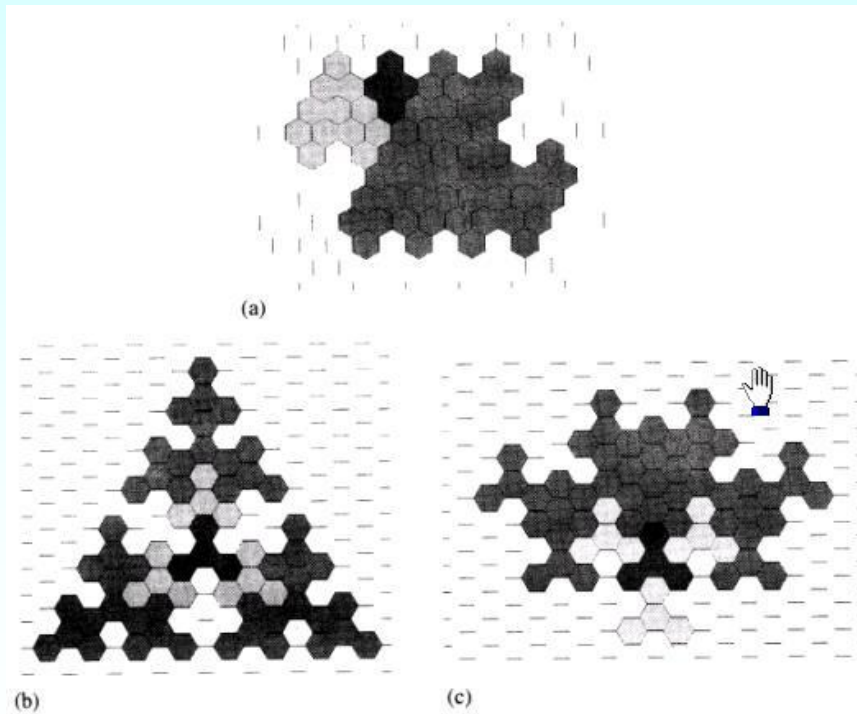
1.5 Mathematic foundation of Raster



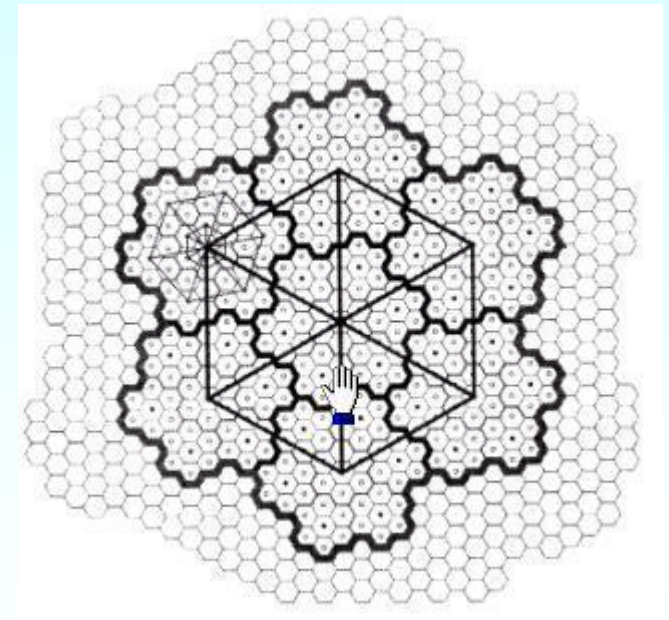
所有单元大小、形状相同且排列一致的等面划分

1.5 Mathematic foundation of Raster

Even with the same kind of equilateral, there remain countless forms of tessellation for the same space.



三种不同的六边形层次划分



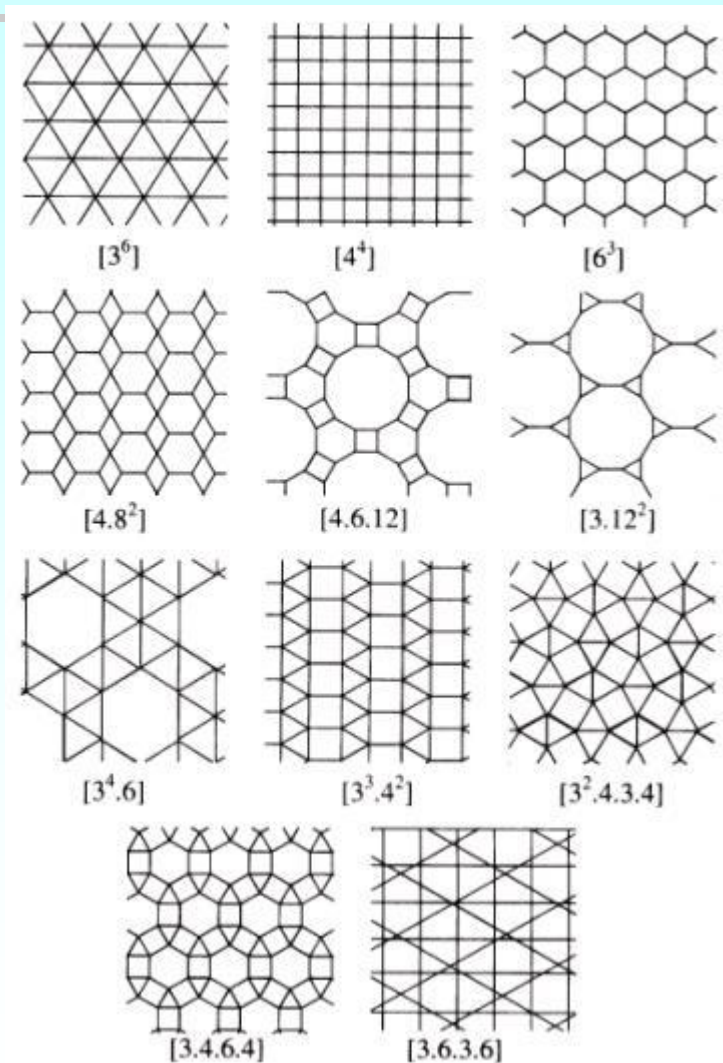
六边形层次划分

1.5 Mathematic foundation of Raster

But if the combined type of the vertexes of all tessellation is limited to be the same, then there will remain 11 kinds of tessellation only. They are named as “Archimedean Tessellation”. 由两种或多种正多边形组成，且每个顶点的正多边形边数排列完全相同（不考虑旋转或反射）。共有**11**种，其中**3**种为正则铺砖（仅一种正多边形），**8**种为半正则铺砖（多种正多边形）。

符号说明

- **数字**：表示正多边形的**边数**（如 3 = 三角形，4 = 正方形，6 = 六边形）。
- **点号分隔**：按顺时针或逆时针顺序列出顶点处的正多边形边数。
- **指数形式**： 3^6 表示 6 个三角形（3.3.3.3.3.3 的简写）。



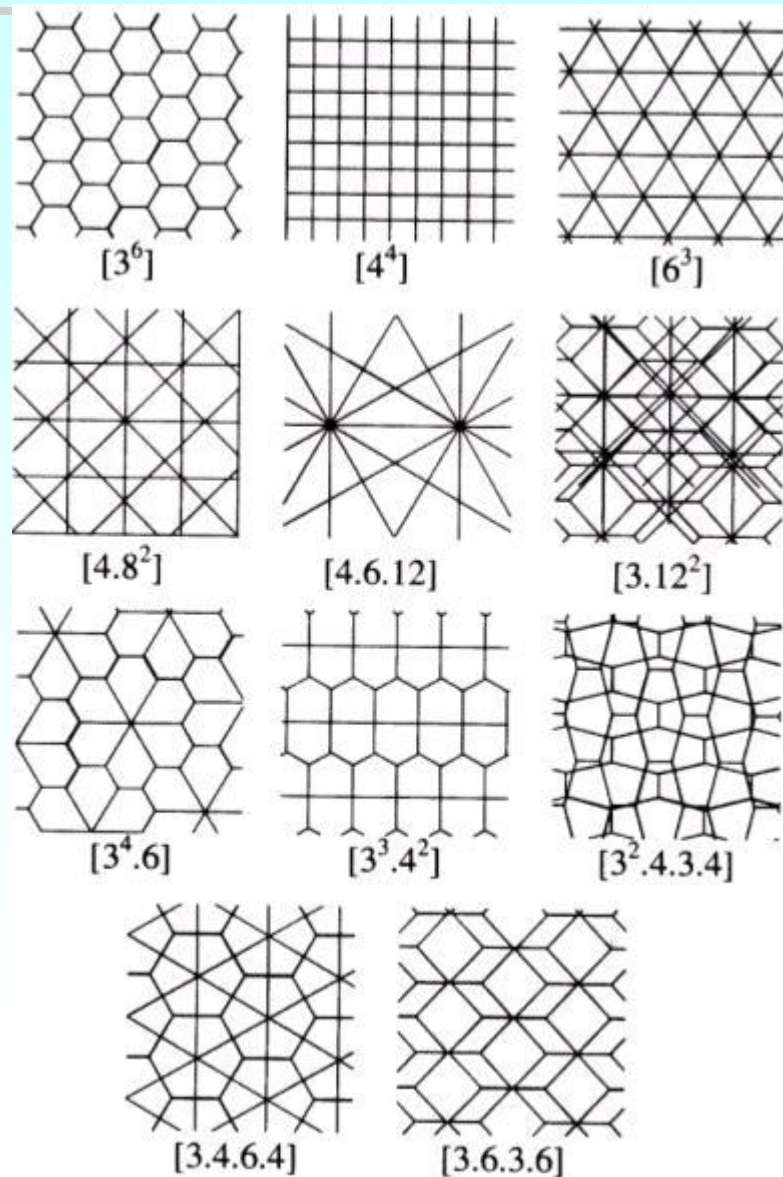
11种不同的阿基米德铺砖（剖分）

1.5 Mathematic foundation of Raster

Another definition concerned about topological relationship between cells is “Laves Tessellation”, which also remains only 11 kinds of tessellation. It is named by Fritz Laves, a famous crystal detector.

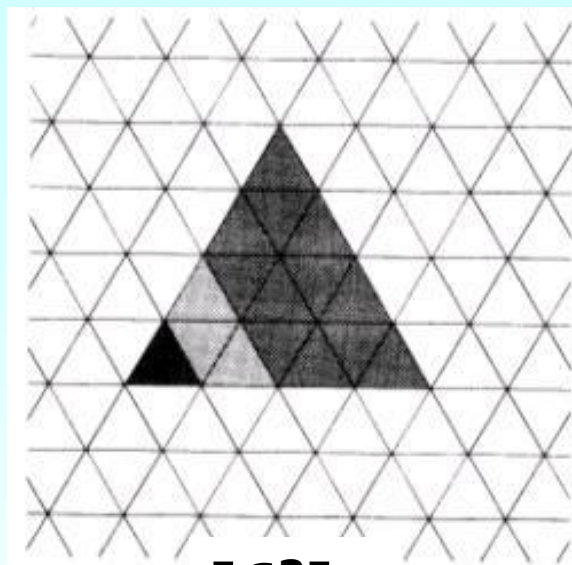
Laves 空间划分方法（Laves tiling 或 Laves partitioning）是一种基于晶体学和几何多面体的三维空间划分方法，其核心思想是通过特定多面体的紧密堆积实现空间的无间隙分割。该方法得名于德国晶体学家Friedrich Laves，他在研究金属间化合物（如 Laves 相）的晶体结构时提出了相关理论。

Laves 三维空间划分方法

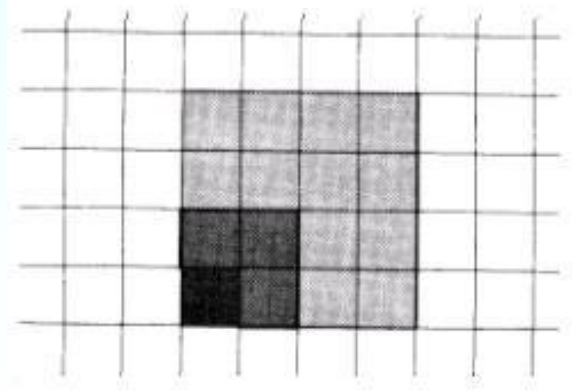


1.5 Mathematic foundation of Raster

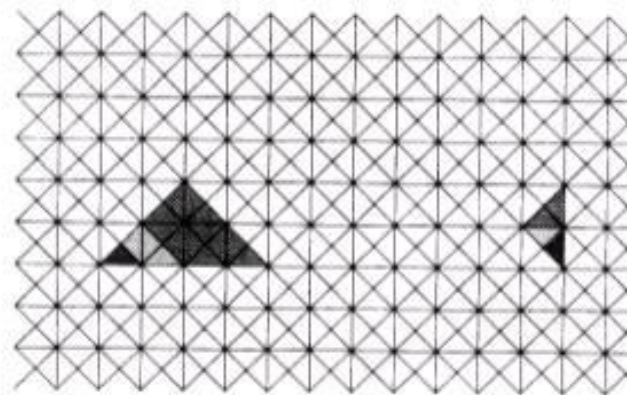
4
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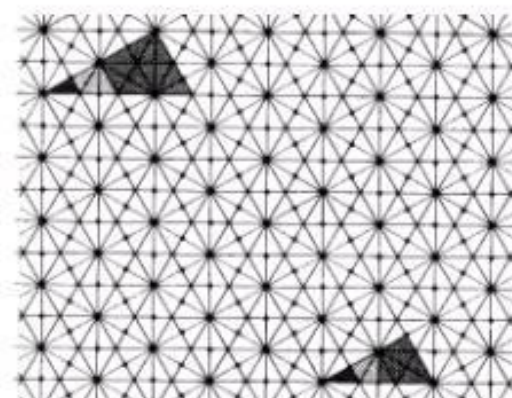
(a) **[6³]**



(b) **[4⁴]**



[4.8²], 普通层次（左）、
旋转层次（右）



[4.6.12], 普通层次
（左），反射层次（右）

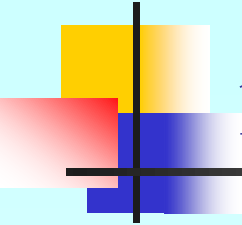
1.5 Mathematic foundation of Raster

宗传明：北京大学数学科学学院教授。主要研究方向为球堆积和数的几何。他彻底解决了二维的深洞问题（由Rogers于1950年提出）；否定了Grunbaum于1961年提出的一个猜想，并且取得了正四面体平移堆积密度（希尔伯特第18问题）的第一个实质性上界，成果入选2014 年中国十大科技成果。



- 17世纪，牛顿与格雷戈里争论的经典问题：一个中心球周围最多能放多少个同尺寸球？
- 牛顿数：在 n 维欧几里得空间中，以一个球为中心，最多可放置 k 个同半径的球，使它们均与中心球表面接触且互不相交，则称 k 为 n 维空间的牛顿数，记作 $K(n)$ 。
- 20 世纪 90 年代，宗传明证明：正四面体的牛顿数为 26，即一个中心正四面体周围最多可紧密排列26 个同尺寸正四面体，且所有四面体仅通过面接触，无重叠。

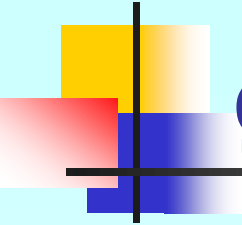
- On the translative packing densities of tetrahedra and cuboctahedra, *Advances in Mathematics*, 260 (2014), 130-190.
- Packing, covering and tiling in two-dimensional spaces, *Expo. Math.* 32 (2014), 297-364.



1.6 Computer Representation of Raster

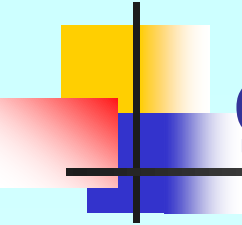
Representing spatial phenomenon with raster data:

- **Digital Terrain Model**
- **Digital Ground (or Geomorphology) Model**
- **Triangulated Irregular Networks**
- **Digital Elevation (Surface) Model, 2.5 dimension**
- **GRID**



1.7 Fundamentals of georeferencing a raster dataset

- The control points are used to build a polynomial transformation that will shift the raster dataset from its existing location to the spatially correct location.
- You have the choice of using a polynomial, a spline, an adjust, or a projective transformation to determine the correct map coordinate location for each cell in the raster.
- The polynomial transformation uses a polynomial built on control points and a least-squares fitting (LSF) algorithm. It is optimized for global accuracy but does not guarantee local accuracy.

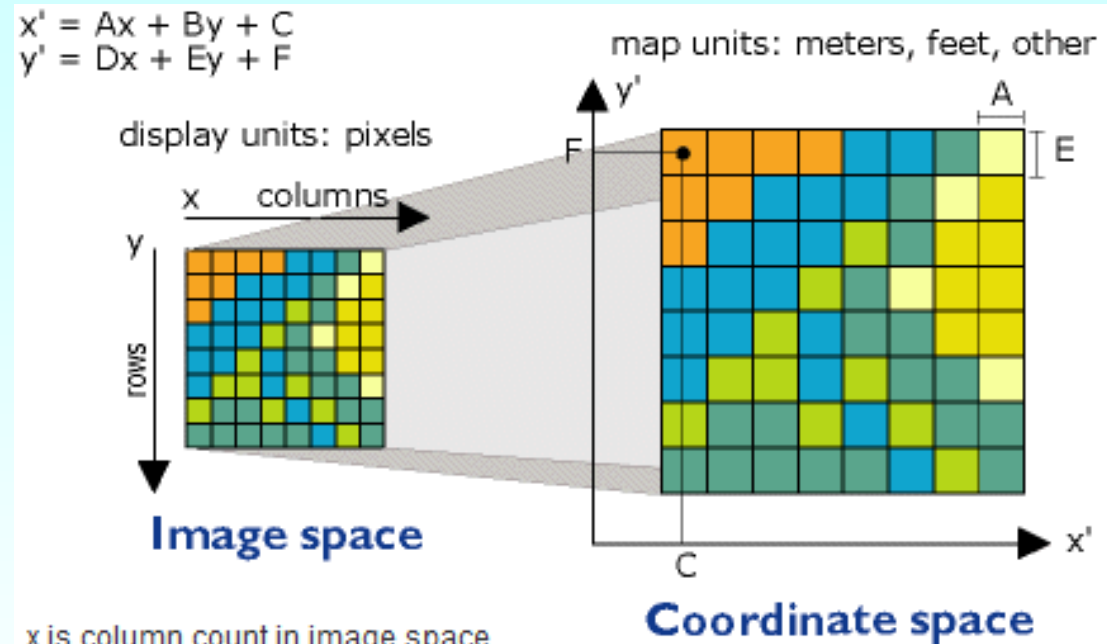


1.7 Fundamentals of georeferencing a raster dataset

- The number of the noncorrelated control points required for this method must be:
 - 1 for a zero-order shift,
 - 3 for a first order affine
 - 6 for a second order
 - and 10 for a third order.
- The lower order polynomials tend to give a random type error, while the higher order polynomials tend to give an extrapolation error.

1.7 Fundamentals of georeferencing a raster dataset

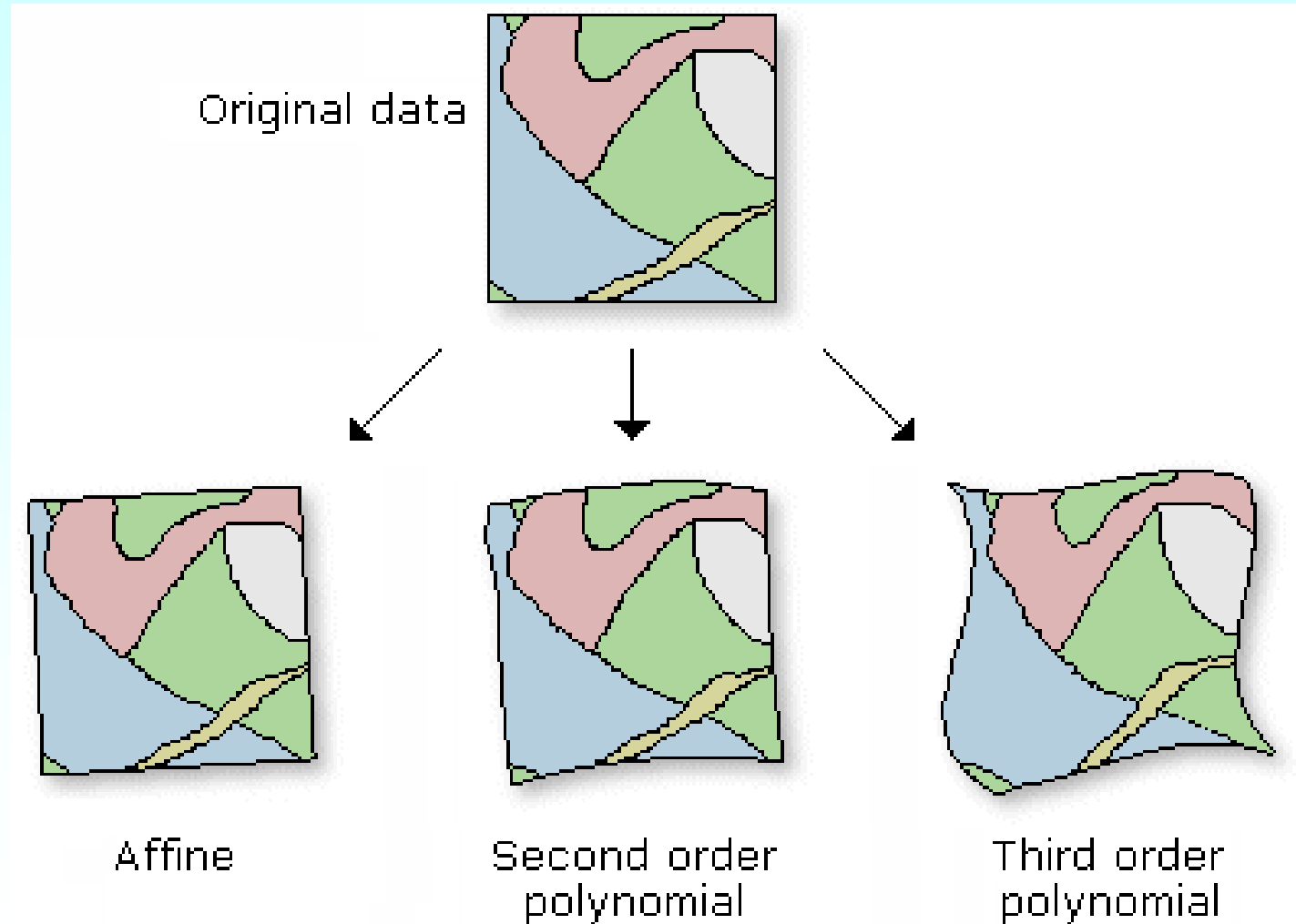
Use a first-order or affine transformation to shift, scale, and rotate a raster dataset.

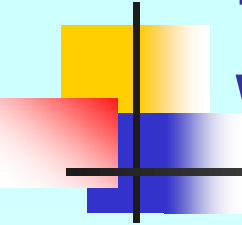


x is column count in image space.
y is row count in image space.
x' is horizontal value in coordinate space.
y' is vertical value in coordinate space.

A is width of cell in map units.
B is a rotation term.
C is the x' value of the center of the upper-left cell.
D is a rotation term.
E is negative height of cell in map units.
F is the y' value of the center of the upper-left cell.

1.7 Fundamentals of georeferencing a raster dataset





2. Some essential problems in SA with Raster

2.1 Euclidean Distance

2.2 Euclidean Direction

2.3 How Flow Direction Works

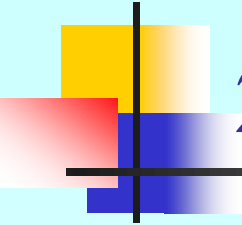
2.4 Window Size and Flow Direction

2.5 Physical Edge or No Data Cells

2.6 Value Attribute Table (关联表)

2.7 Output setting of value

2.8 Fundamental operations

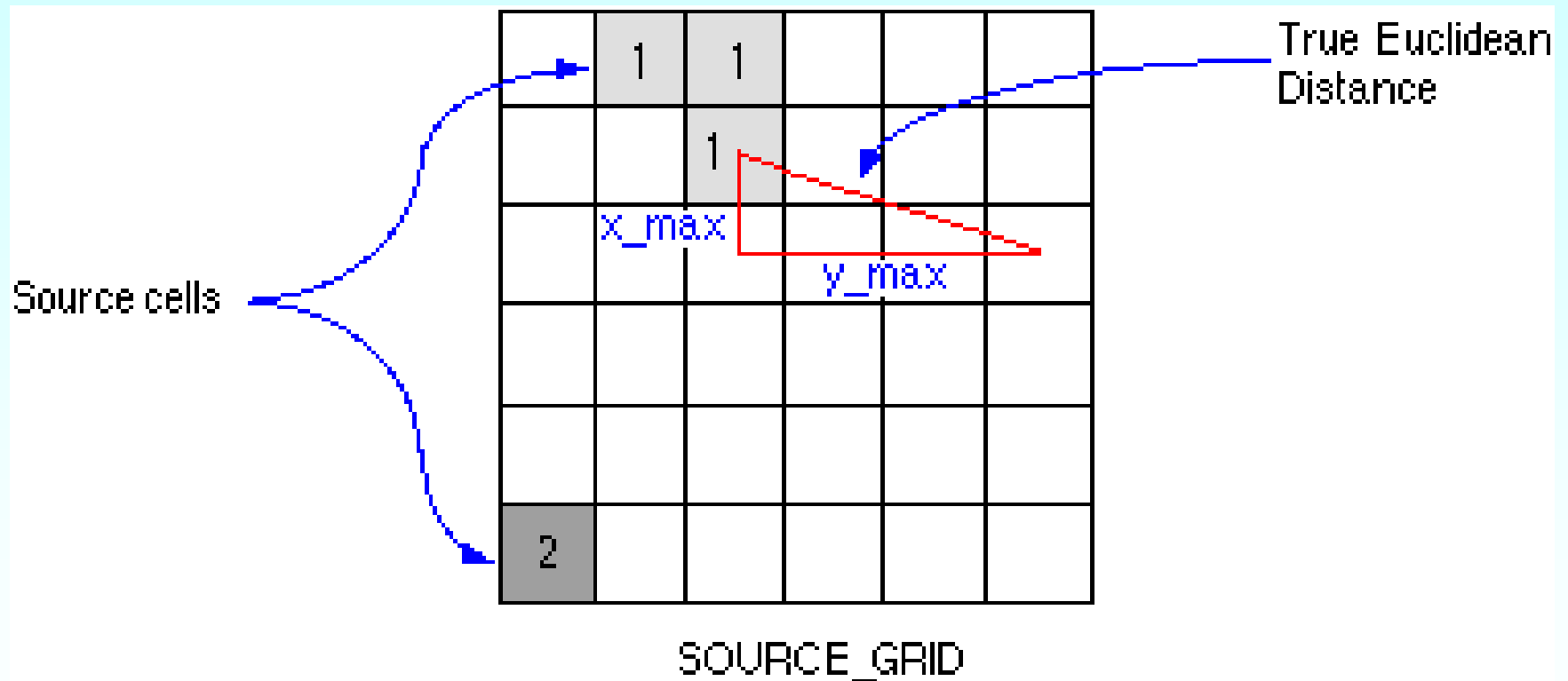


2.1 Euclidean Distance

Euclidean distance is calculated from the center of the source cells to the center of each of the surrounding cells.

Conceptually, the Euclidean algorithm works as follows. For each cell, the distance is calculated to each source cell by calculating the hypotenuse with the x-max and y-max as the other two legs of the triangle. This calculation derives the true Euclidean, not cell, distance. The shortest distance to a source is determined and if it is less than the specified maximum distance, the value is assigned to the cell location on the output raster.

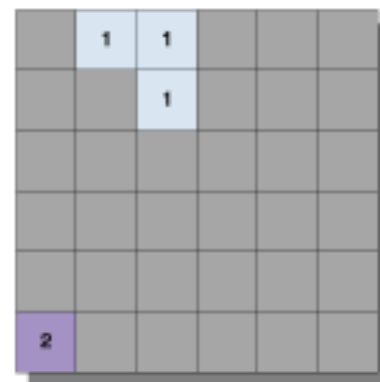
2.1 Euclidean Distance



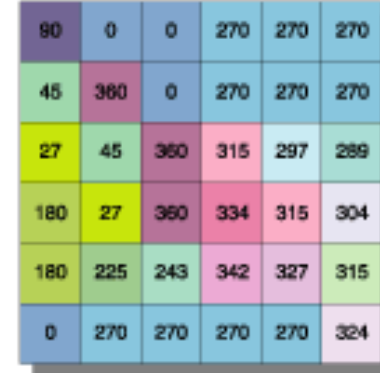
2.2 Euclidean Direction

Calculates the direction in degrees that each cell center is from the cell center of the closest source.

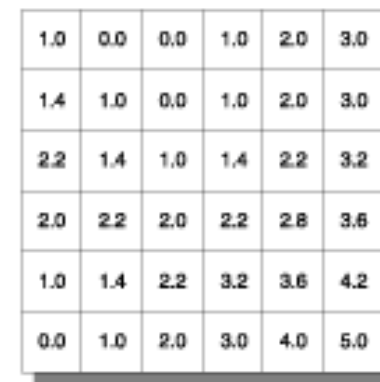
The output values are based on compass directions, with 0° being reserved for the source cells (90° to the east, 180° to the south, 270° to the west, and 360° to the north).



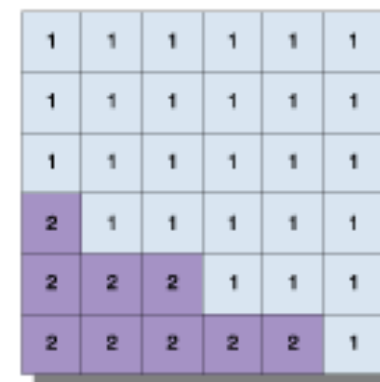
SOURCE_GRID



EUC_DIRECTION_GRID



EUC_DISTANCE_GRID

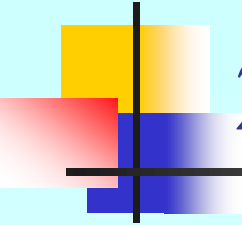


EUC_ALLOCATION_GRID



VALUE=NODATA

Expression: EUCDIRECTION(SOURCE_GRID,
EUC_DISTANCE_GRID, EUC_ALLOCATION_GRID)



2.3 How Flow Direction Works

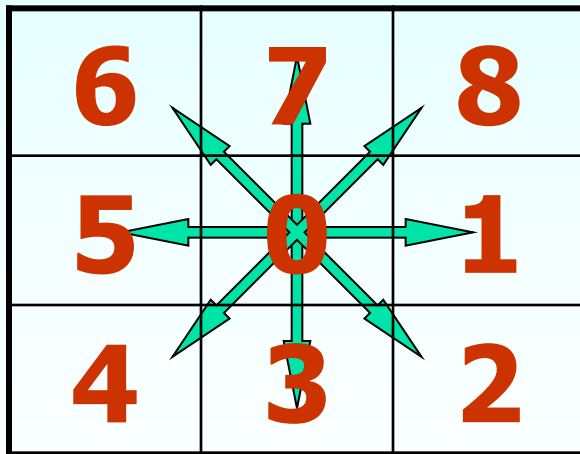
- The direction of flow is determined by finding the direction of steepest descent from each cell. This is calculated as

$$\text{change in z value} / \text{distance} * 100$$

- The distance is determined between cell centers. Therefore, if the cell size is 1, the distance between two orthogonal cells is 1, and the distance between two diagonal cells is 1.414.
- If the descent to all adjacent cells is the same, the neighborhood is enlarged until a steepest descent is found.
- If all neighbors are higher than the processing cell, it will be considered noise and filled to the lowest value of its neighbors, and have a flow direction towards this cell. However, if a one-cell sink is next to the physical edge of the raster or has at least one NoData cell as a neighbor, then it is not filled due to insufficient neighbor information. To be considered a true one-cell sink, all neighbor information must be present.

2.4 Window Size and Flow Direction

Cost-Weighted Direction



30	90	24
18	124	75
15	5	

3	4	3
2	2	3
1	1	0

2.4 Window Size and Flow Direction

Creates a raster of flow direction from each cell to its steepest downslope neighbor.

direction definition: The output of Flow Direction is an integer raster whose values range from 1 to 255.

78	72	69	71	58	49
74	67	58	49	46	50
69	53	44	37	38	48
64	58	55	22	31	24
68	61	47	21	16	19
74	53	34	12	11	12

ELEVATION

=

2	2	2	4	4	8
2	2	2	4	4	8
1	1	2	4	8	4
128	128	1	2	4	8
2	2	1	4	4	4
1	1	1	1	4	16

FLOW_DIR

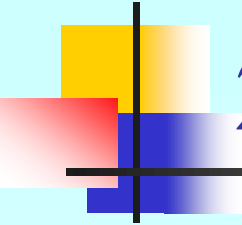


VALUE=NODATA

32	64	128
16		1
8	4	2

Expression: FLOWDIRECTION(ELEVATION)

128编码比8编码有何优势？



2.4 Window Size and Flow Direction

- Using the **2ⁿ** number to identify some cells that can not be determined about the direction.
- If a cell has the same change in z value in multiple directions and that cell is part of a **sink**, the value for that cell in the output flow-direction raster will be the sum of those directions. For example, if the change in z value is the same, both to the right (flow direction = 1) and down (flow direction = 4), the flow direction for that cell is $1 + 4 = 5$.
- **Cells with undefined flow direction can be flagged as sinks using the Sink function.**

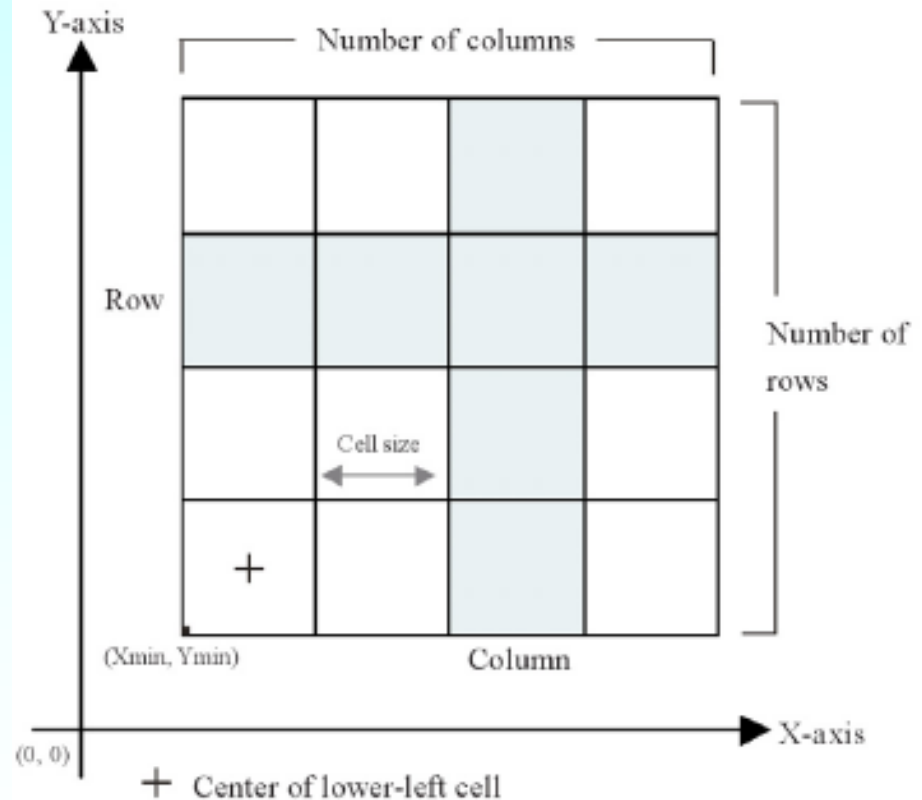
2.5 Physical Edge or No Data Cells

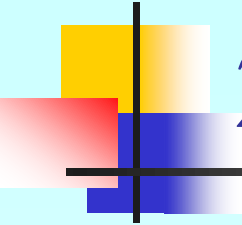
1	1	0	0	0	1	1	0
	1	2	2	3	3	1	2
4	0	0	2		0	0	2
4	0	1	1	3	2	1	0

GRID



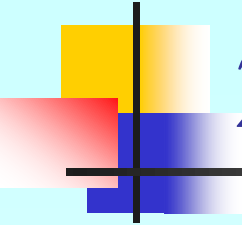
VALUE=NODATA





2.5 Physical Edge or No Data Cells

Cell values can be either positive or negative, integer, or floating point. Cells can also have a NoData value to represent the absence of data. Sometimes there are homogeneous areas in a raster dataset that the user does not want to display. These can include borders, backgrounds, or other data considered to not have valid values. Sometimes these are expressed as NoData values, although other times they may have real values.



2.5 Physical Edge or No Data Cells

1. 在一个运算符或局部函数中、邻域函数的邻域中或分区函数的分类区中，如果任何输入栅格的某个位置上存在空值，则输出单元位置上也为空值。

2. 忽略空值单元并用所有有效值完成计算。

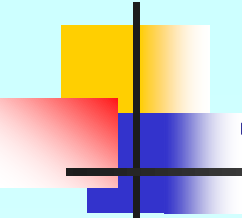
当在两个栅格数据集之间或对局部函数应用操作符时，第二种方式（忽略了空值）并不可行。当空值单元存在于邻域函数中某单元的邻域内或分区函数的分类区中时，缺省情况下，可以计算所有已知值的单元的和（**sum**）、中值（**median**）、种类（**Variety**）、众数（**majority**）、寡数（**minority**）并分配给输出栅格数据集（此缺省值可被覆盖）。

2.6 Value Attribute Table (关联表)

栅格数据集关联的属性表

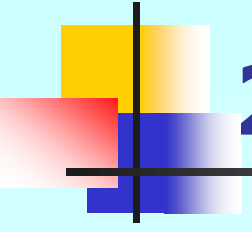
- 第一项是值 (Value)
- 第二项是个计数 (Count)
- Value和Count都是强制性字段

VEGETATION VAT				
VALUE	COUNT	TYPE	CANOPY	BUG-DAM
1	12	Maple	30	8
2	8	Oak	65	10
3	4	Field	0	0
4	5	Hickory	45	20
5	4	Pine	80	35



2.7 Output setting of value

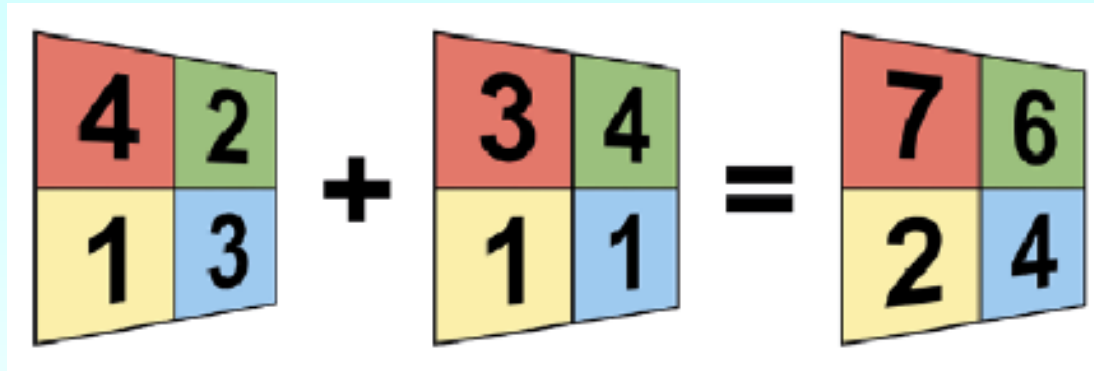
Bit depth	Range of values that each cell can contain
1 bit	0 to 1
2 bit	0 to 4
4 bit	0 to 16
Unsigned 8 bit	0 to 255
Signed 8 bit	-128 to 127
Unsigned 16 bit	0 to 65535
Signed 16 bit	-32768 to 32767
Unsigned 32 bit	0 to 4294967295
Signed 32 bit	-2147483648 to 2147483647
Floating-point 32 bit	-3.402823466e+38 to 3.402823466e+38



2.8 Fundamental operations

- **Mathematical operations and functions**
- **Bitwise** manipulation
- **Logical evaluation**
- **Trigonometric calculations**

2.8 Fundamental operations



A diagram illustrating the addition of two 2x2 grids. The first grid has values 4 (red), 2 (green), 1 (yellow), and 3 (blue). The second grid has values 3 (red), 4 (green), 1 (yellow), and 1 (blue). The result grid has values 7 (red), 6 (green), 2 (yellow), and 4 (blue). The operation is represented by a plus sign and an equals sign.

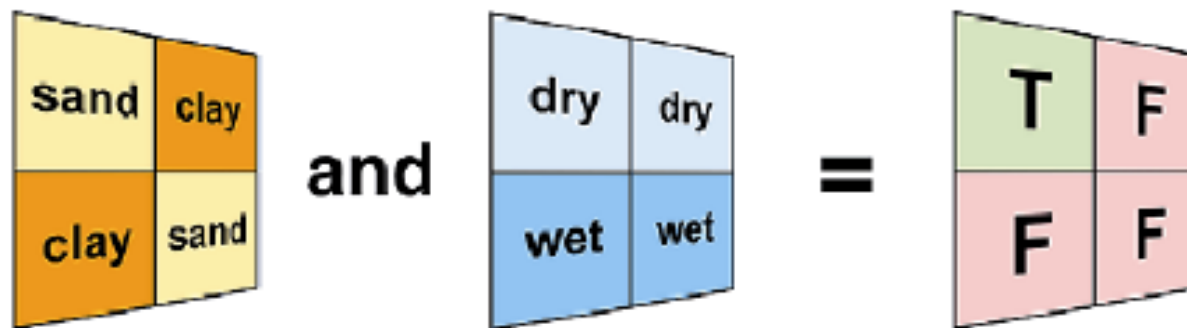
4	2
1	3

 +

3	4
1	1

 =

7	6
2	4



A diagram illustrating the logical AND operation on two 2x2 grids. The first grid contains the words 'sand' and 'clay'. The second grid contains the words 'dry' and 'wet'. The result grid contains the truth values 'T' (True) and 'F' (False). The operation is represented by the word 'and' and an equals sign.

sand	clay
clay	sand

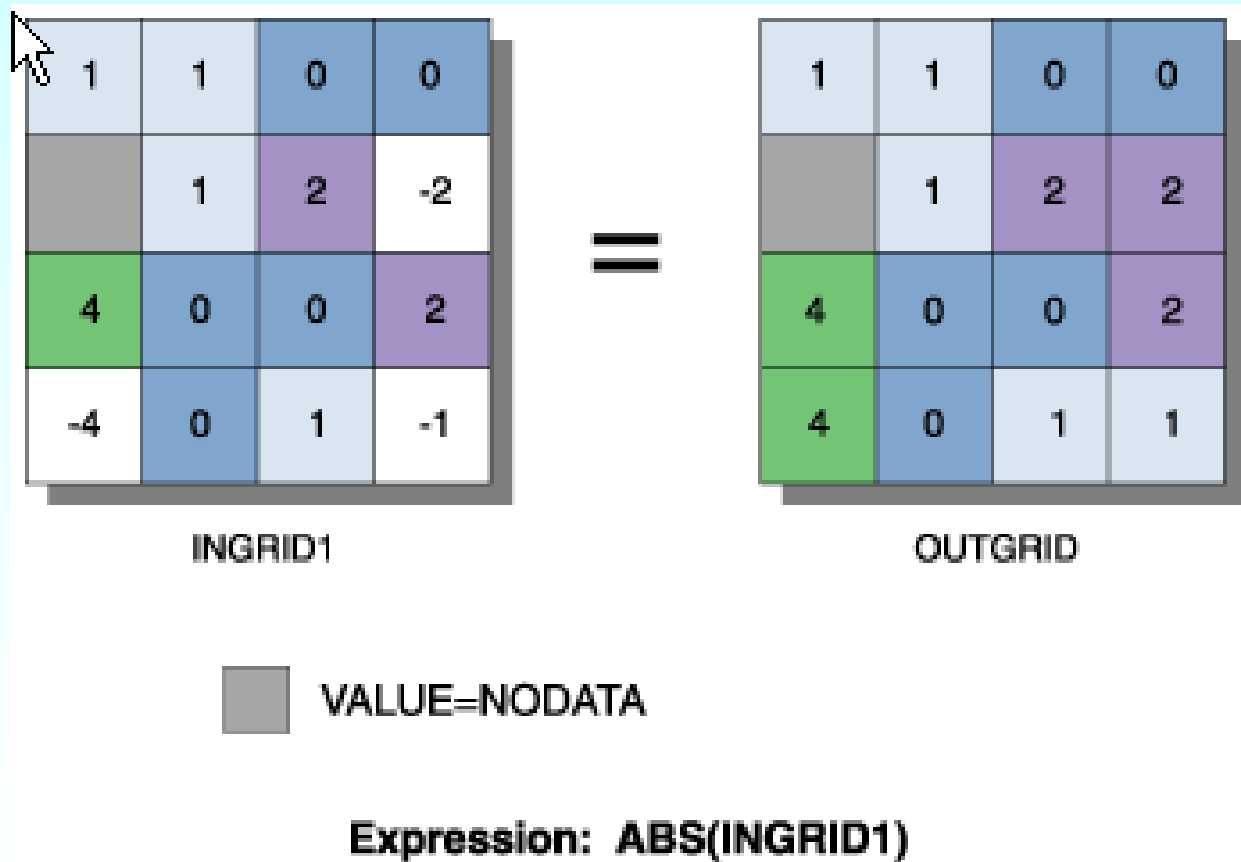
 and

dry	dry
wet	wet

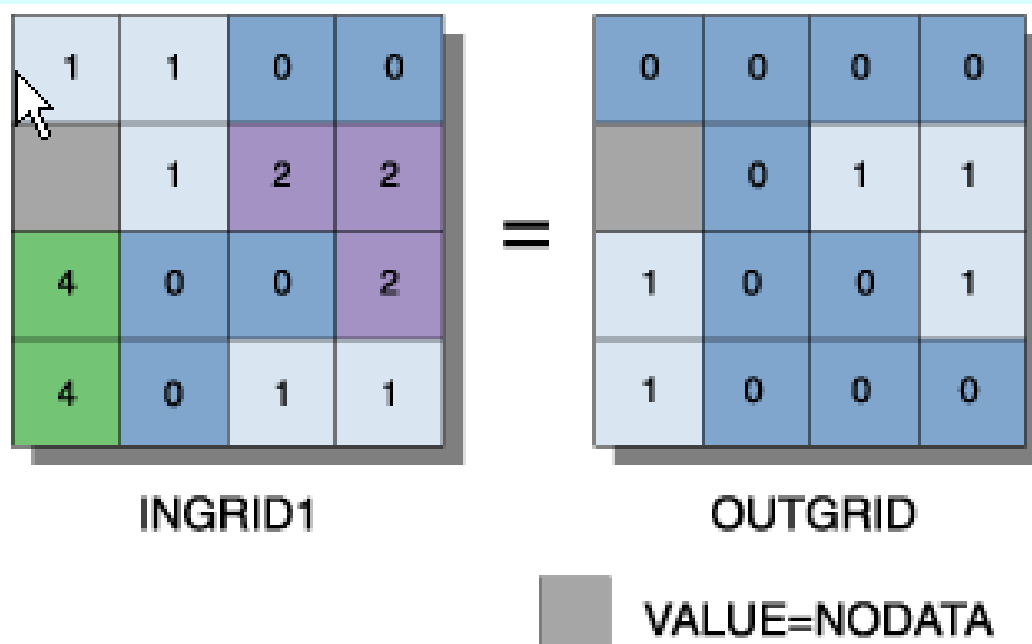
 =

T	F
F	F

2.8 Fundamental operations

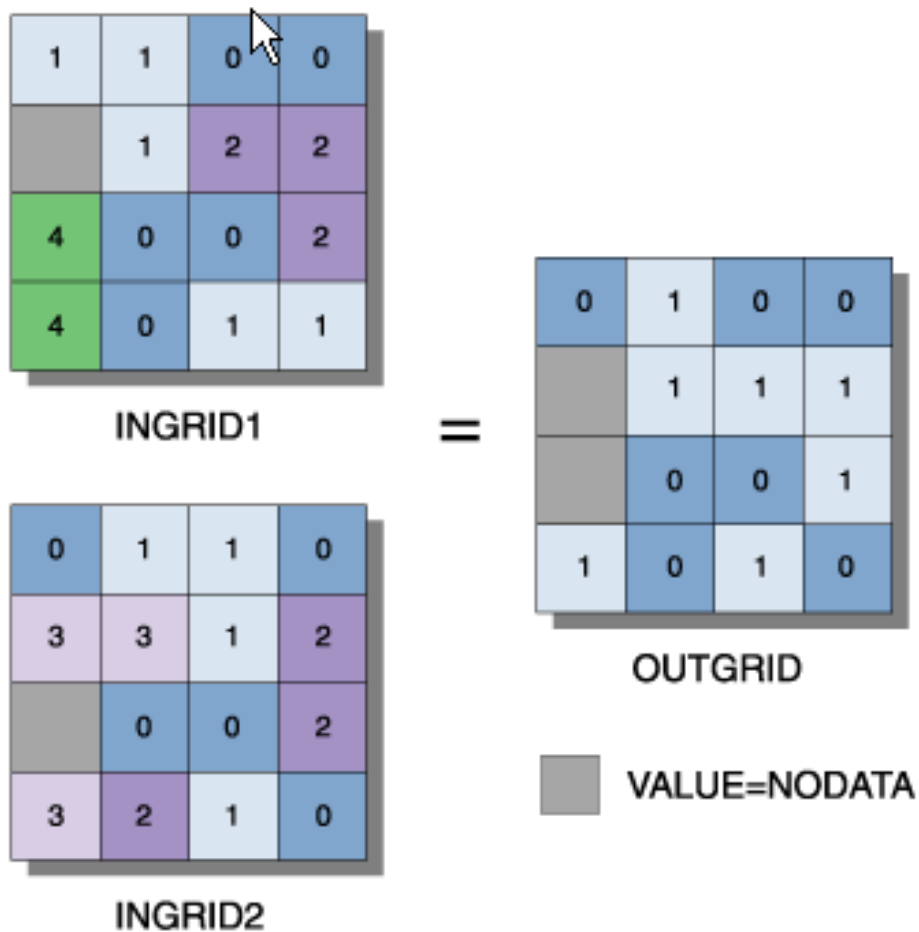


2.8 Fundamental operations

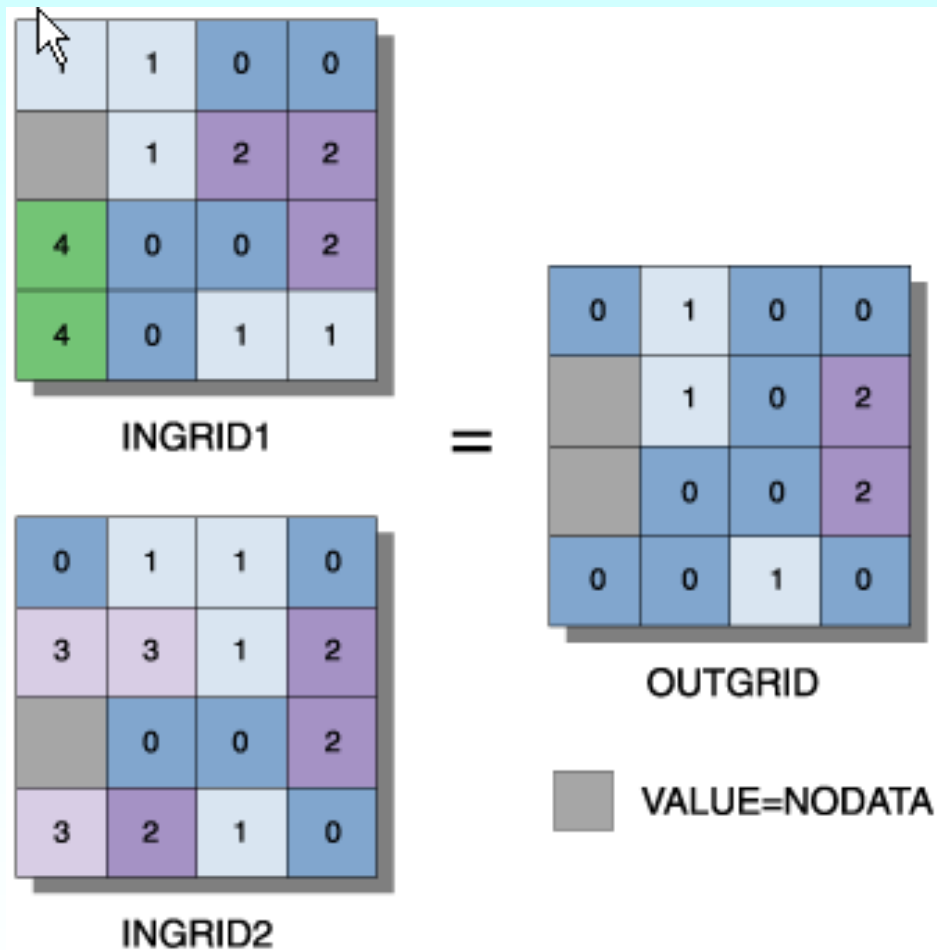


Expression: **TEST(INGRID1, 'VALUE >= 2')**

2.8 Fundamental operations



Expression:
 $OUTGRID = INGRID1 \& INGRID2$



Expression:
 $OUTGRID = INGRID1 \&\& INGRID2$

2.8 Fundamental operations

1	1	0	0
	1	2	2
4	0	0	2
4	0	1	1

INGRID1

OUTGRID

VALUE=NODATA

Expression: SIN(INGRID1)

0	1.0	2.0	10.0
30.0	45.0	57.29	60.0
90.0	180.0	270.0	360.0
-45.0	-0.5	0.5	

INGRID1

OUTGRID

VALUE = NODATA

Expression: SIN(INGRID1 DIV DEG)

2.8 Fundamental operations

□ 变化检测

- 以逐个像元比较的方式，确定第一个输入的哪些值与第二个输入的值在逻辑上不同。
- 如果两个输入的值不同，则使用第一个输入的值作为输出。如果两个输入的值相同，则输出为 0。

1	1	0	0
	1	2	2
4	0	0	2
4	0	1	1

InRas1

0	1	1	0
3	3	1	2
	0	0	2
3	2	1	0

InRas2

=

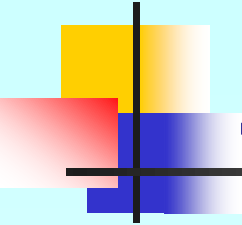
1	0	0	0
	1	2	0
	0	0	0
4	0	0	1

OutRas

用途举例？

■ 值 = NoData

$$\text{OutRas} = \text{Diff}(\text{InRas1}, \text{InRas2})$$



2.8 Fundamental operations

- 空间分析/地图代数/栅格计算器中的其他函数

- InList

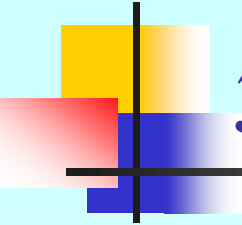
- Pick

- IsNull

- Over

- Test

-



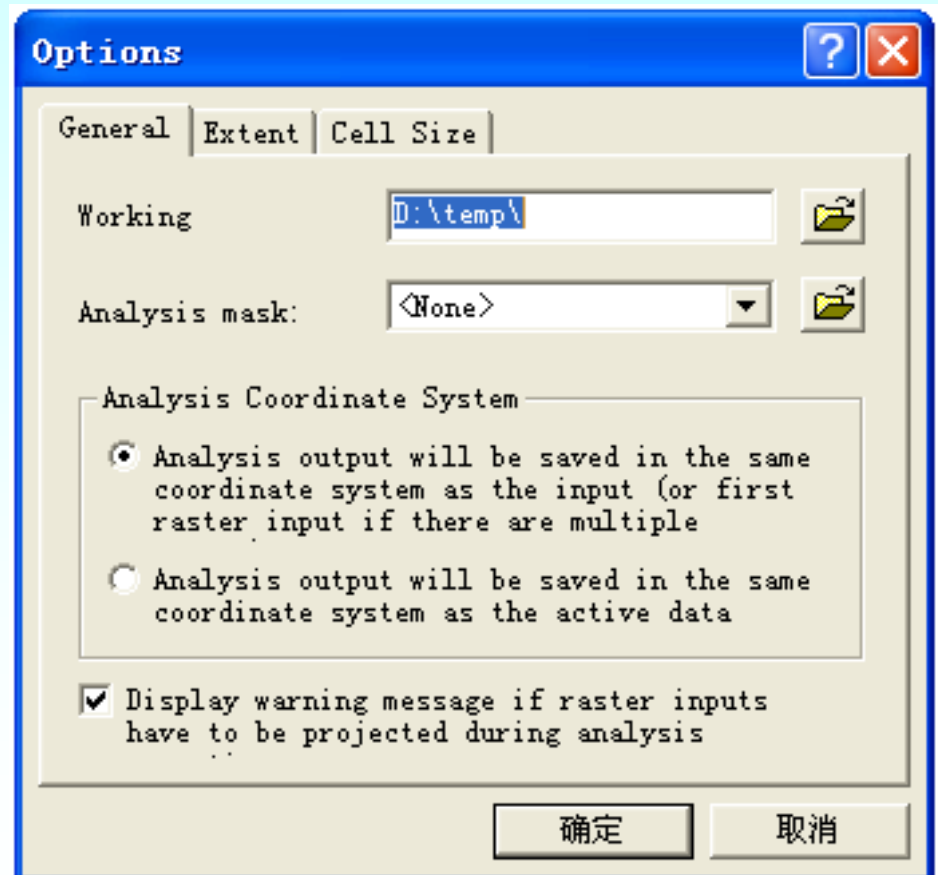
3. Function Framework

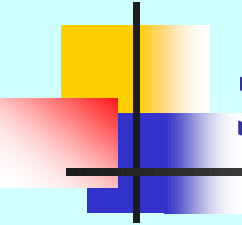
1. Distance Mapping
2. Density Mapping: Contour line drawing
3. Location Based Modeling
4. Map Algebra: image processing and integrating analysis
5. Surface Analysis: Extraction for terrain factor
6. 3-Dimension Analysis
7.

3.1 Setup for Analysis

- Working Path/Directory
- Analysis Mask/Extent
- Coordinate system
- process file management
- distance direction definition
- Source definition
- Cost definition
- *Maximum distance*
- Output cellsize
- Cost distance weighted data
- allocation data

Spatial Analyst → Option



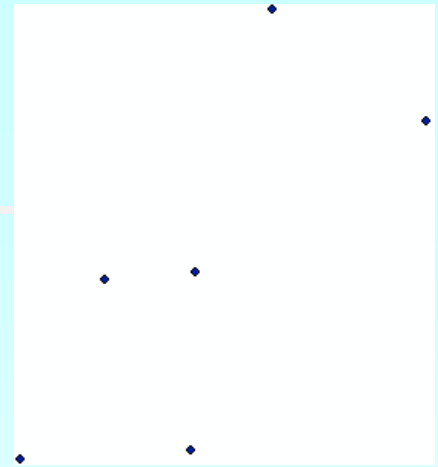


3.2 Line Distance

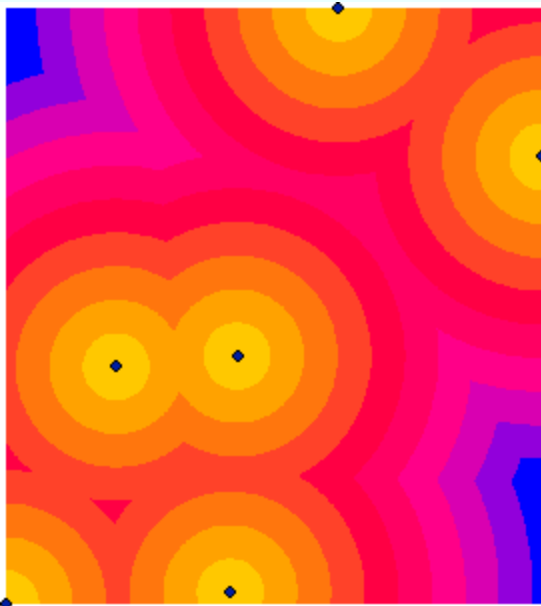
- Calculate the distance of each cell to its nearest sink.
- Application in air pollution.
- input: point layer
- output:
 - Direction Raster
 - Allocation Raster
 - Distance Raster (Output Raster)

3.2 Line Distance

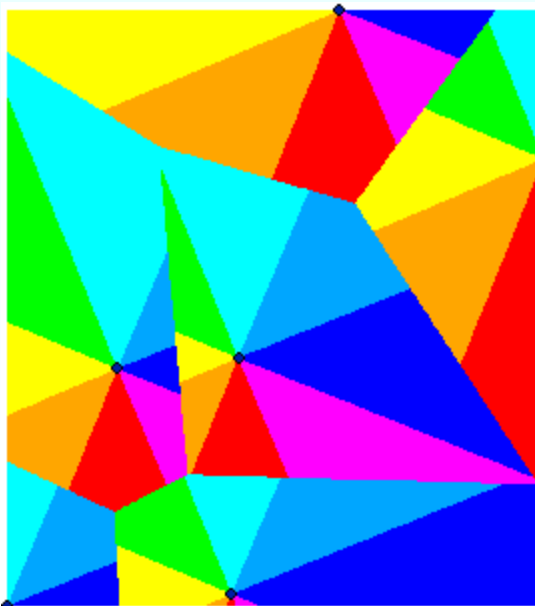
input



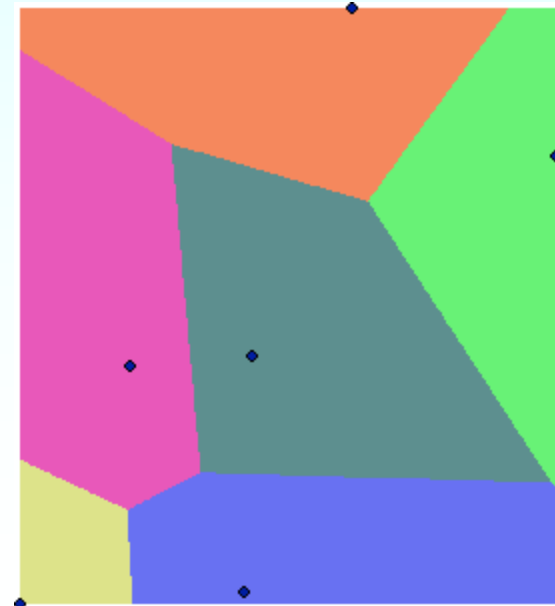
Distance

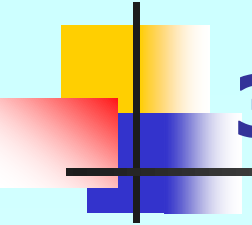


Output
Direction



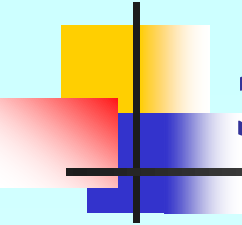
Allocation





3.3 Region Allocation

- Allocate all cells to their nearest sink. The identity of the sink was assigned to the cells.
- To find out the serve area of supermarkets or schools.
 - Input :point layer
 - Output: Allocation Raster

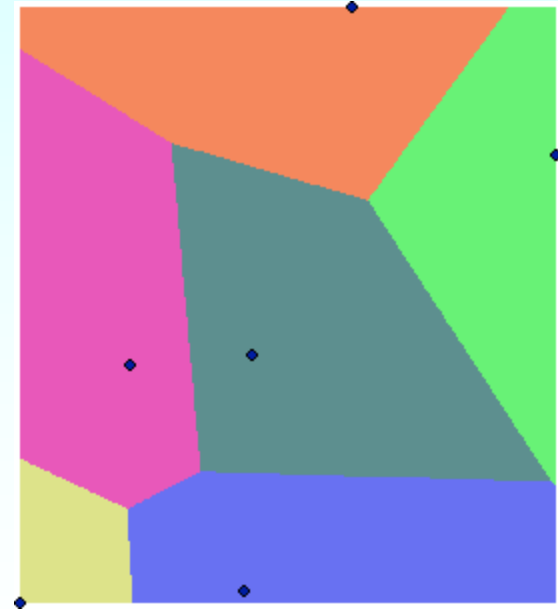


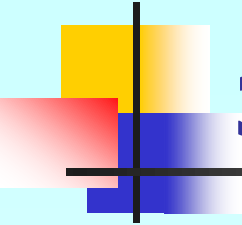
3.3 Region Allocation

input



output

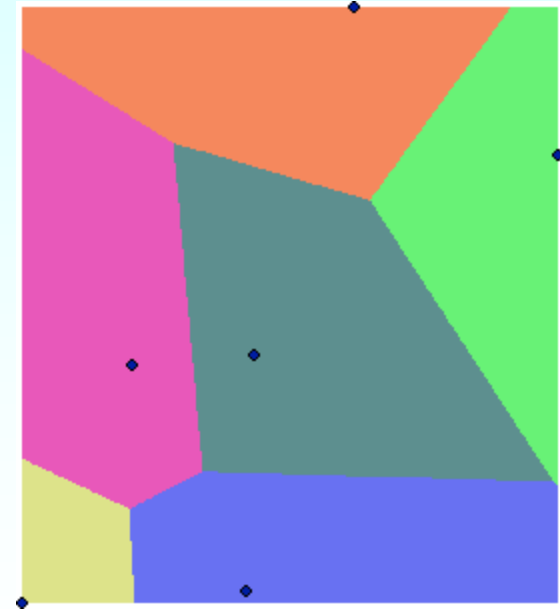
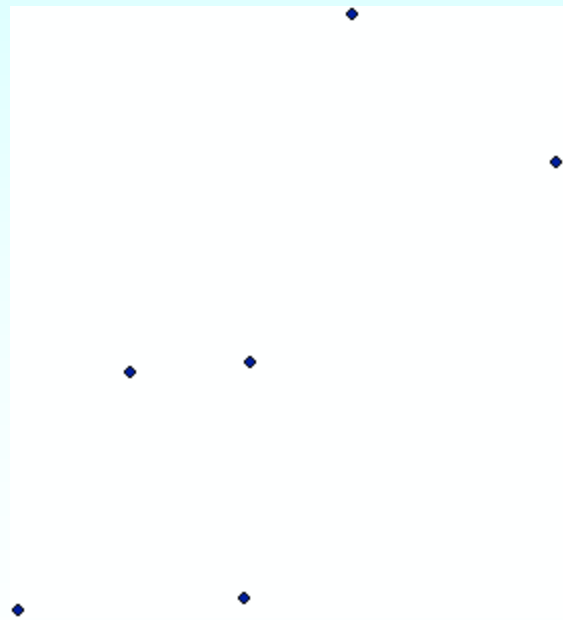


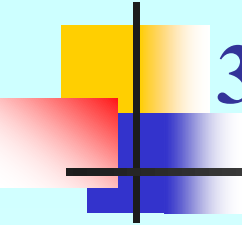


3.3 Region Allocation

Question:

How to plot Theissen polygon?





3.4 Cost Distance Weighted

- 计算出每个栅格到距离最近、成本最低源的最少累加成本。
- 同时可生成两个相关输出：
 - 成本方向数据：从每一单元出发，沿着最低累计成本路径到达最近源的具体路线。
 - 成本分配数据：记录每个单元的隶属源信息。
- 输入：
 - 源，点层
 - 成本权重，Raster层
- 输出：
 - Direction Raster
 - Allocation Raster
 - Distance Raster (Output Raster)

3.4 Cost Distance Weighted

parameter setting for cost distance weighted



Cost Weighted

Distance to: center of city

Cost raster: denggx_10.img

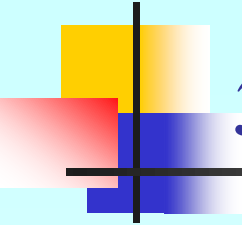
Maximum distance:

☒ Create direction: D:\temp\dir_cw1

☒ Create allocation: D:\temp\al_cw1

Output raster: D:\temp\cw1

OK Cancel

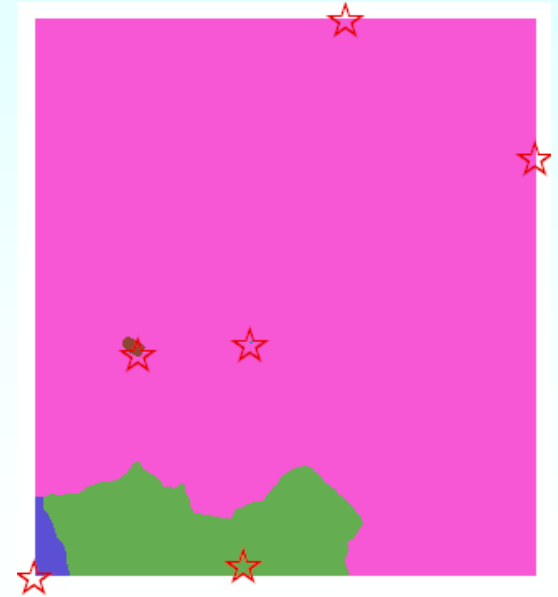
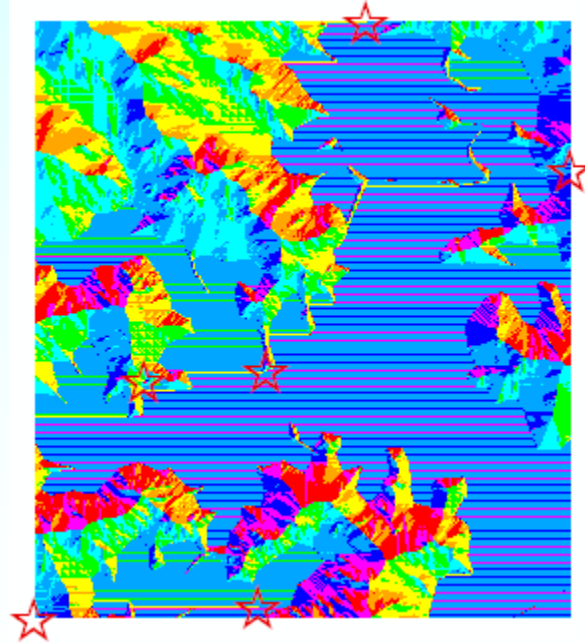
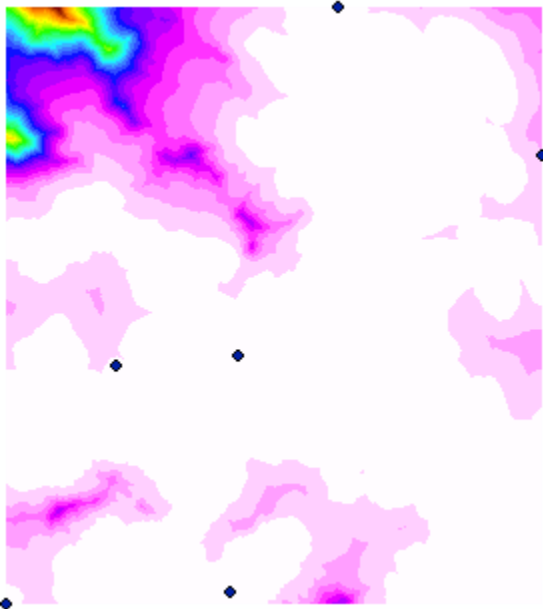


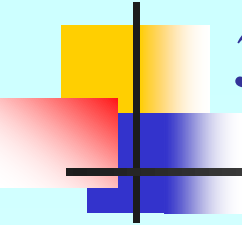
3.4 Cost Distance Weighted

5	2	3	3					
4	3	2	3					
2	①	2	2					
3	2	4	4					
1	3	5	⑥					
0	2	3	4					
Cost_Raster				Cost_Distance				

3.4 Cost Distance Weighted

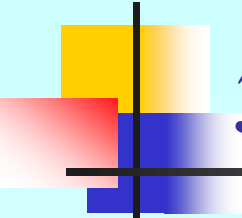
cost distance weighted





3.5 The Shortest Route

- 从一个源或一组源出发，到达一个目的地或一组目的地的最短直线路径或最小成本路径。
- 对应源的不同，有三种最短路径计算方法：
 - **For Each Cell:** 为源中每一个单元点寻找一条成本最小路径。
 - **For Each Zone:** 为每个源寻找一条成本最小路径，源中所有单元共享同一条路径。
 - **For Each Zone Best Single:** 为所有源寻找一条成本最小路径，此时，只有一个源与一个相应的目标点或目标组相连。



3.5 The Shortest Route

path 多指由人或动物在田间、小丘或树丛中经常走动而踩出的小路，也指公园或花园等供人走的小径。

road 指连接两地间供行人或车辆使用的广阔平坦的大道，多指公路，也可用于引申意义。

way 普通用词，含义广泛，可指各种路、道或通道，也可指抽象的道路。

street 尤指城市中的道路，往往一侧或两侧有高大建筑物；还可指城市小镇、近郊可供人、车、马行走的平坦道路。

highway 通常指市区外可以通行各种机动车辆的交通干线。

avenue 在美国，指城市中一侧或两侧建筑物林立的大道或马路；而在英国则常指通往乡村大住宅、私人大庄园或两旁栽树的道路。

motorway 高速公路，新词，特指为车辆快速通行而修的专线，美语为**freeway**或**expressway**。

route 指从此处通往彼处的路线、路程等。

lane 指农村或城镇的小道或小径，也指小巷。

pavement 在英国，多指街道两旁的人行道，而美国人常用**sidewlk**表示人行道。

trail 指人或兽在森林、荒野或山中踩出的小径或崎岖小道。

3.5 The Shortest Route

Shortest Path [?] [X]

Path to: center of city [v] [icon]

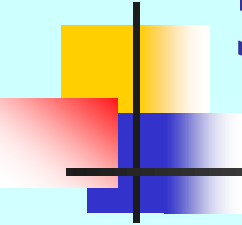
Cost distance raster: denggx_10.img [v] [icon]

Cost direction raster: denggx_10.img [v] [icon]

Path type: For Each Cell [v]

Output features:
For Each Cell
For Each Zone
Best Single

OK Cancel

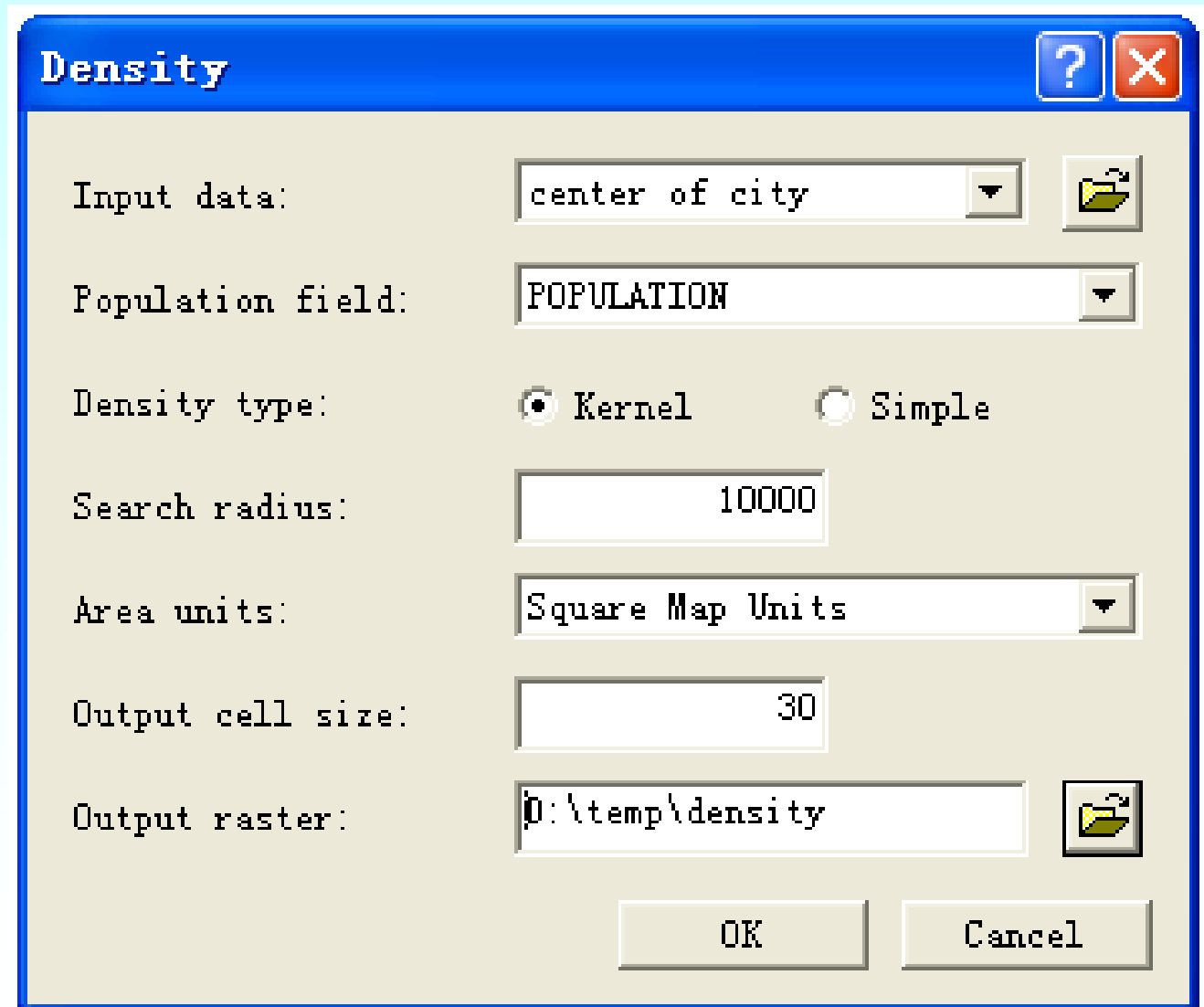


3.6 Density Mapping

- 根据输入的要素数据集计算整个区域的数据聚集状况，从而产生一个连续的密度表面。密度制图主要是基于点数据生成的，以每个待计算格网点为中心，进行圆形区域搜寻，进而来计算每个格网点的密度值。
- 这是一个通过离散采样点进行表面内插的过程。
 - 核函数密度制图**Kernel**: 靠近格网搜寻区域中心的点或线会被赋予较大的权重，随着其与格网中心距离的加大，权重降低。计算结果分布较平滑。
 - 简单密度制图**Simple**: 落在搜寻区域内的点或线有同样的权重，先对其进行求和，再除以搜索区域的面积，从而得到每个点的密度值。
- 应用：如计算人口空间分布密度。

3.6 Density Mapping

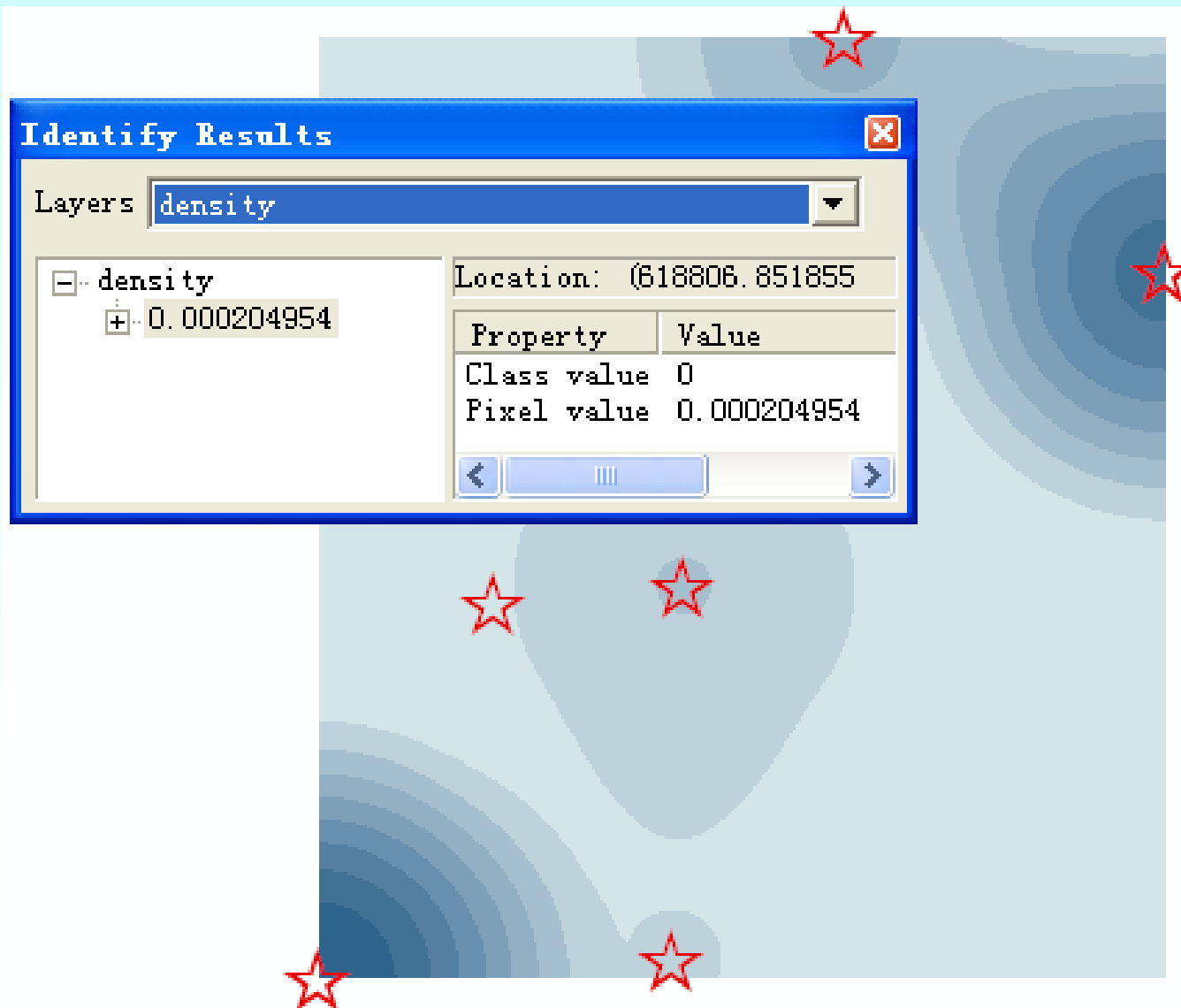
density mapping
setting

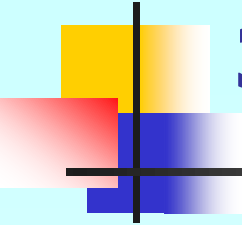


The image shows a 'Density' dialog box with a blue title bar and standard window controls (help, close). It contains several input fields and options for configuring a density map. The 'Input data' field is set to 'center of city' with a browse button. The 'Population field' is set to 'POPULATION'. The 'Density type' has two radio buttons: 'Kernel' (selected) and 'Simple'. The 'Search radius' is set to 10000. The 'Area units' are set to 'Square Map Units'. The 'Output cell size' is set to 30. The 'Output raster' is set to 'D:\temp\density' with a browse button. At the bottom are 'OK' and 'Cancel' buttons.

Input data:	center of city	
Population field:	POPULATION	
Density type:	☒ Kernel ☐ Simple	
Search radius:	10000	
Area units:	Square Map Units	
Output cell size:	30	
Output raster:	D:\temp\density	
OK Cancel		

3.6 Density Mapping





3.6 Density Mapping

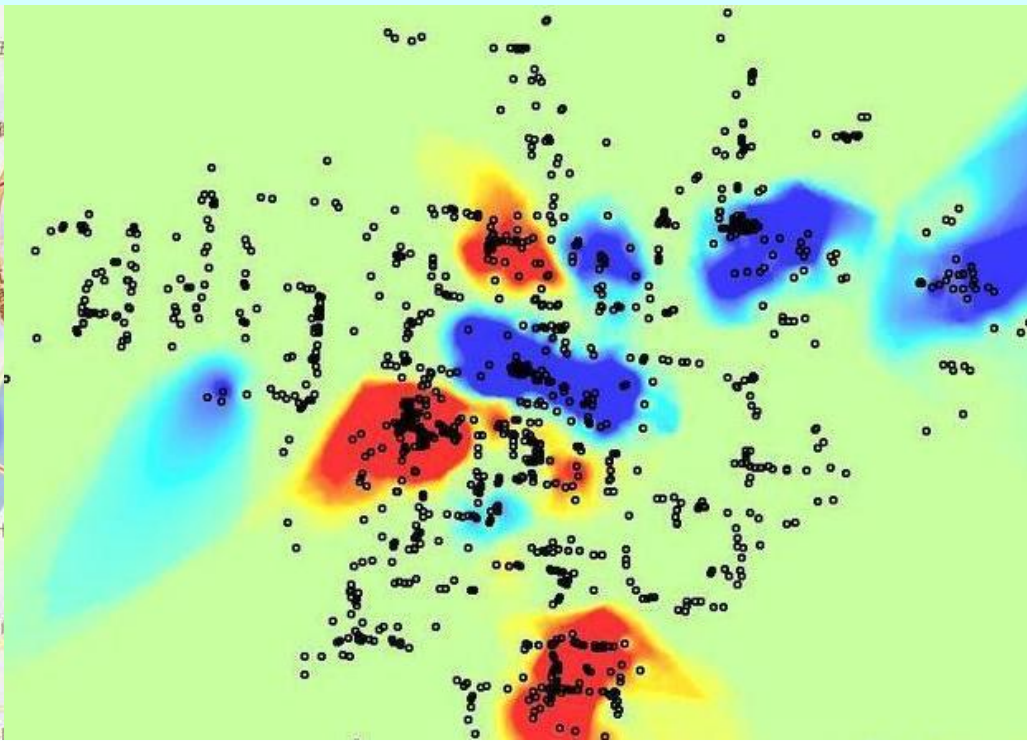
思考问题：

1. 热点图（Hot Spot Map）与热力图（Heat Map）有何区别？
<https://www.gislounge.com/heat-maps-in-gis/>;
<https://www.gislounge.com/difference-heat-map-hot-spot-map/>
2. 点密度图与空间插值有何区别？
3. 想象一下，人口密度的意义是什么？
4. 如何计算种群密度（如老鼠的密度，蚯蚓的密度）？样方法；标志重捕法
5. 如果点的大小（或权重）不同，点密度图如何计算？有何意义？
6. 河网密度、路网密度是常见的统计指标，如何计算？有何意义？
7. 根据计算公式，密度计算受哪些因素影响？空间尺度效应

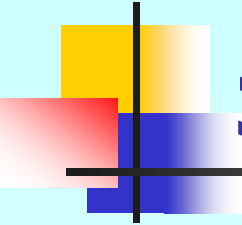
3.6 Density Mapping



热力图



热点图



3.7 Surface Analysis

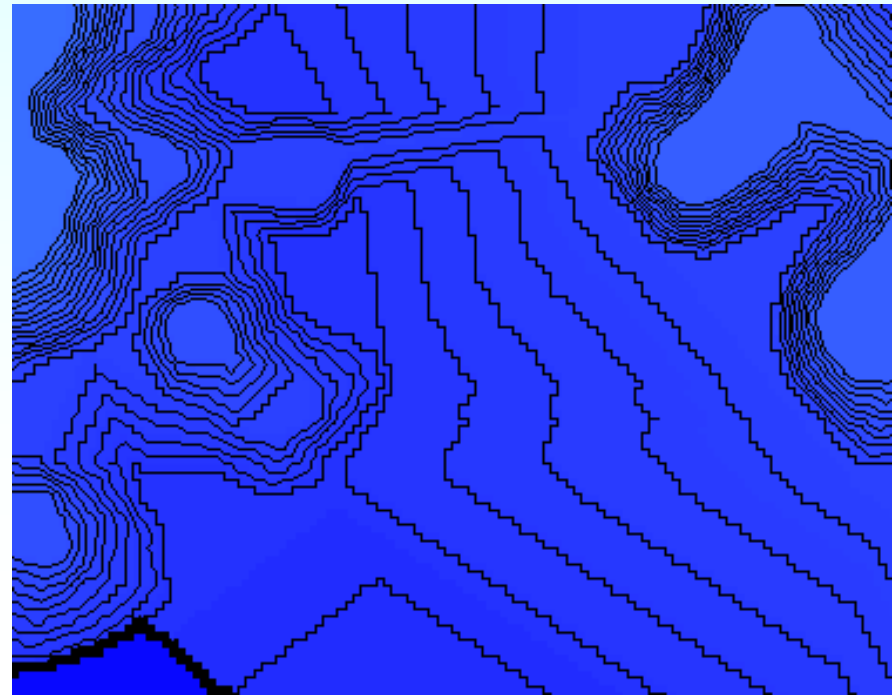
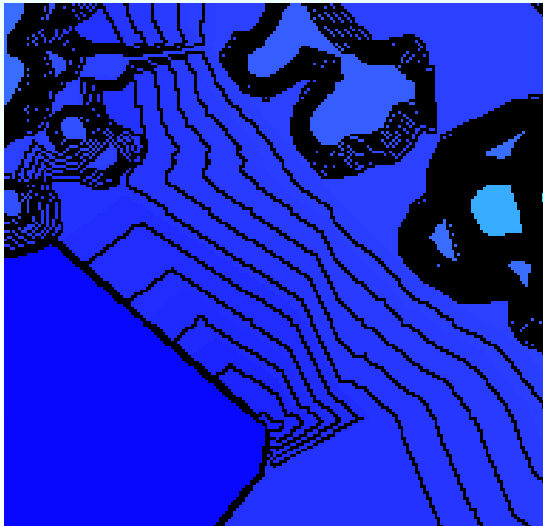
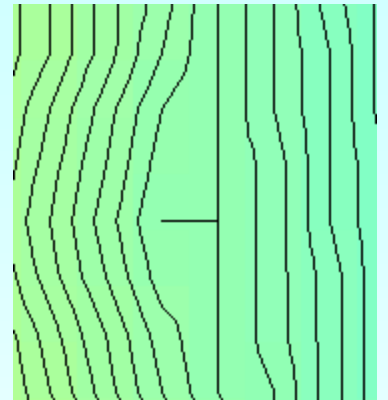
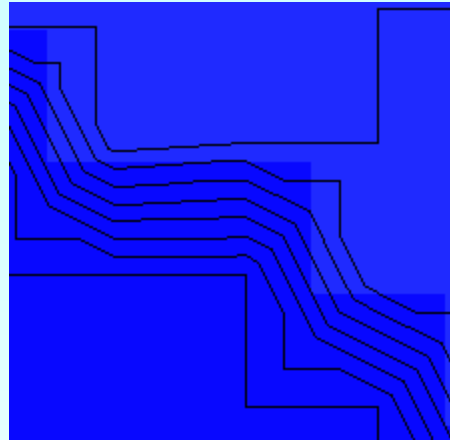
- 栅格插值：IDW、SPLINE、KRIGING
- 数据重采样：ArcToolBox → Data Management Tools → Raster → Resample
- 等值线绘制
- 地形因子提取：坡度、坡向、平面曲率、剖面曲率
- 山体阴影

3.7.1 Contour line

为什么会出现这些奇形怪状的
等值（高）线？如何解释？

original interval: 10—50M

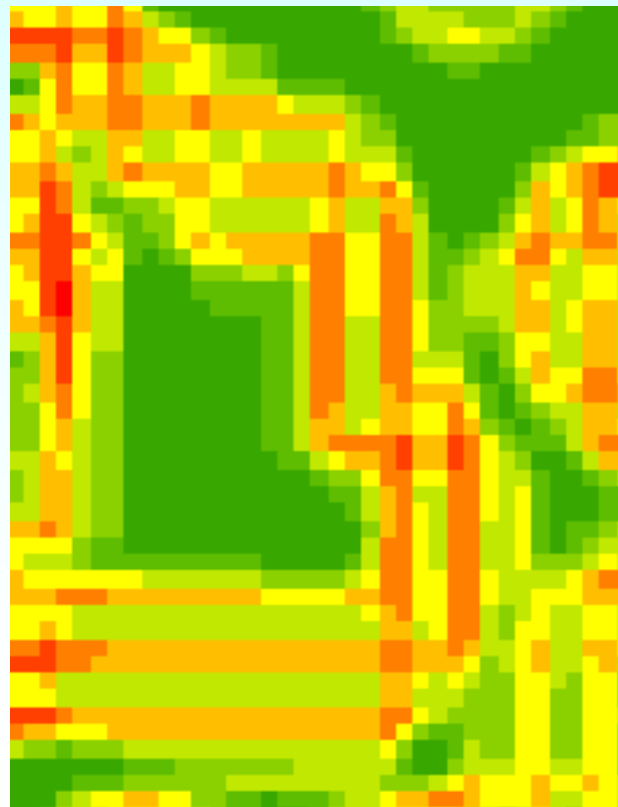
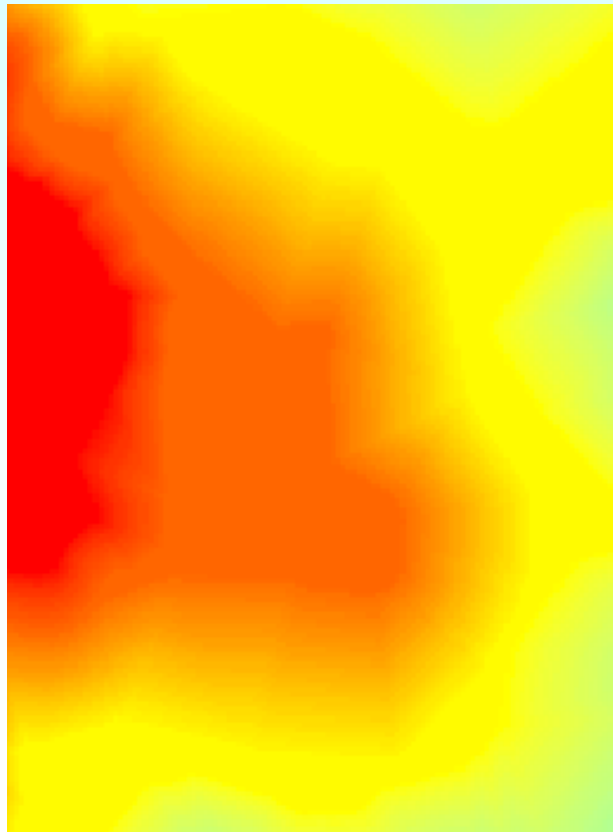
Current interval: 2M



3.7.2 Slope map

坡度的解释与异常：陡坎与悬崖(cliff)会

出现什么情况？ Resolution:30m

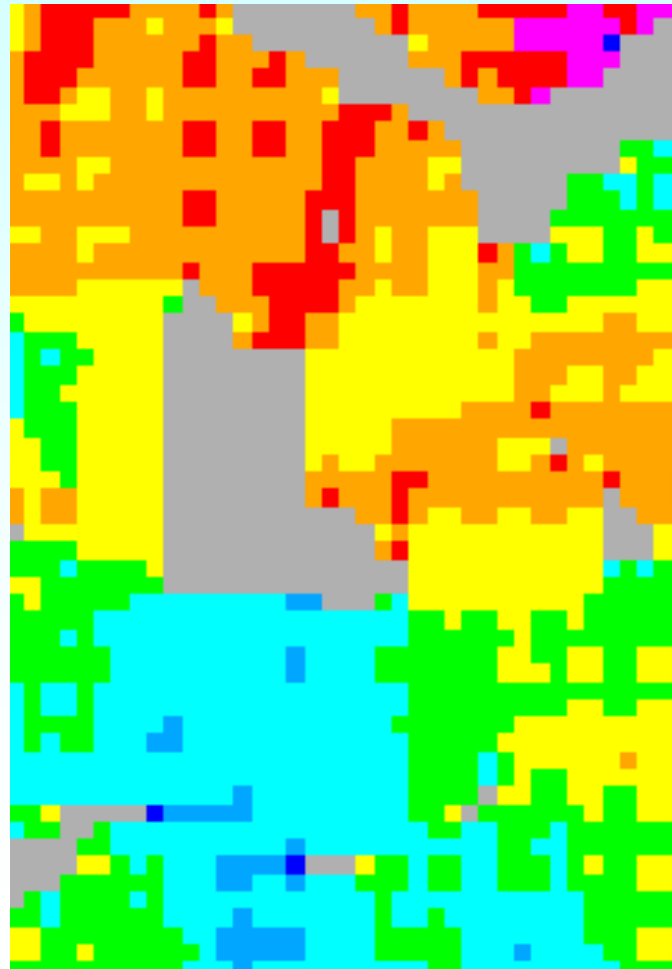
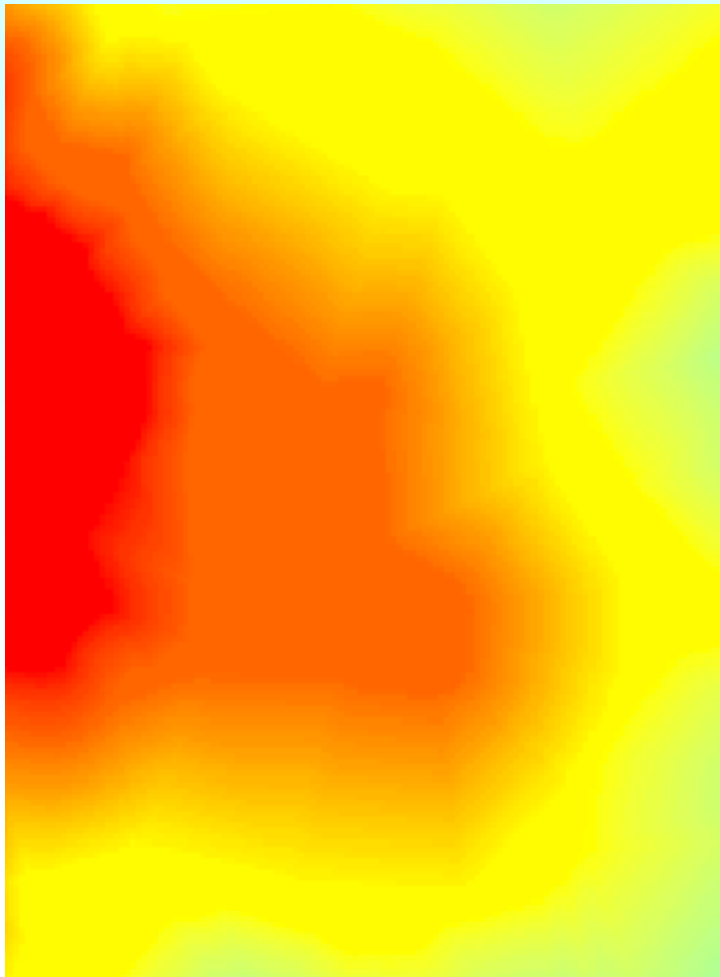
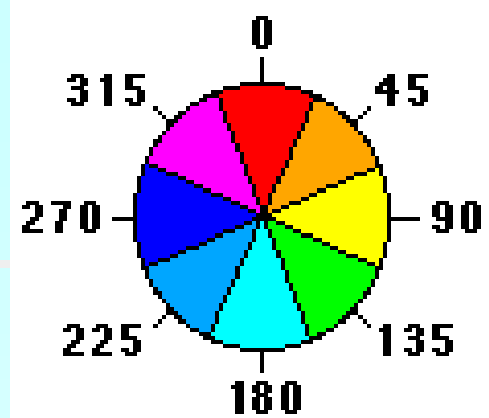


slope1

0 - 1.367036372
1.367036373 - 5.012466699
5.0124667 - 9.569254607
9.569254608 - 14.12604252
14.12604253 - 18.68283042
18.68283043 - 23.23961833
23.23961834 - 28.02424563
28.02424564 - 33.94806992
33.94806993 - 58.32688522

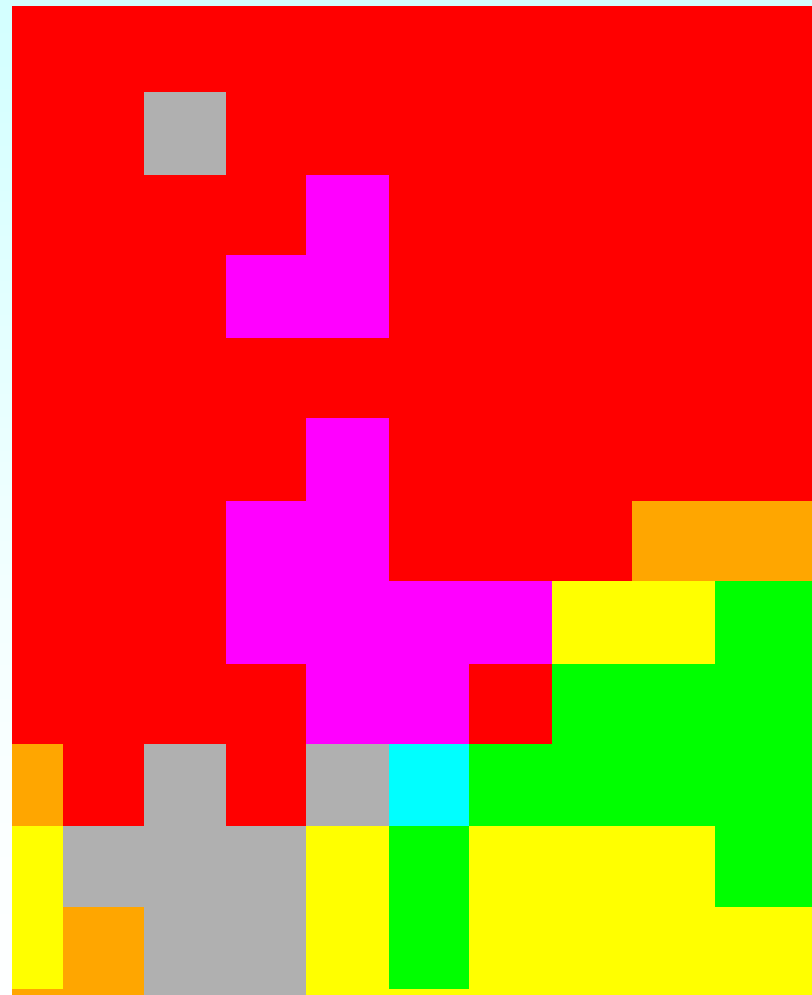
3.7.3 aspect map

坡向的解释



3.7.3 aspect map

some strange
phenomena
in aspect
map.



aspect

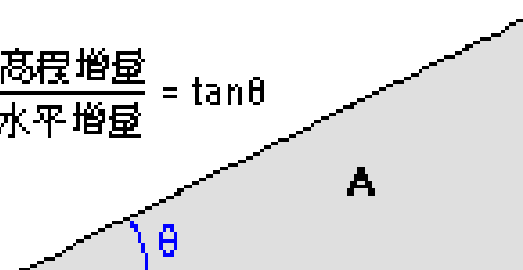
- Flat (-1)
- North (0-22.5)
- Northeast (22.5-67.5)
- East (67.5-112.5)
- Southeast (112.5-157.5)
- South (157.5-202.5)
- Southwest (202.5-247.5)
- West (247.5-292.5)
- Northwest (292.5-337.5)
- North (337.5-360)

3.7.4 Slop Calculation

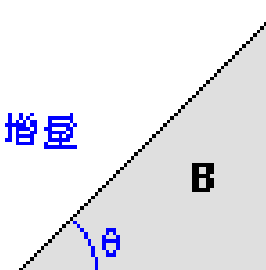
坡度 = θ

坡度百分比 = $\frac{\text{高程增量}}{\text{水平增量}} * 100$

$$\frac{\text{高程增量}}{\text{水平增量}} = \tan\theta$$



高程增量



水平增量

坡度 =

坡度百分比 =

30

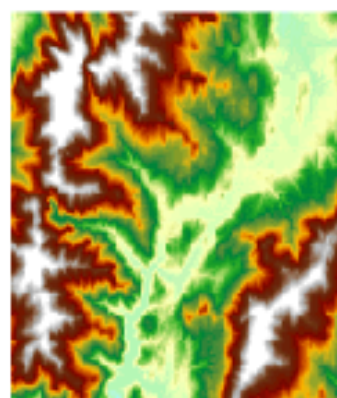
58

45

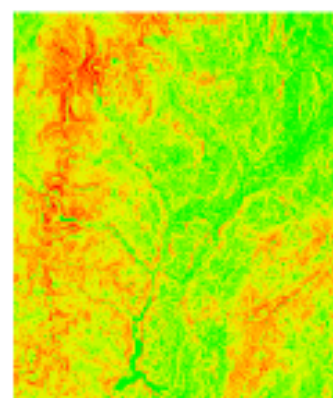
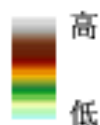
100

76

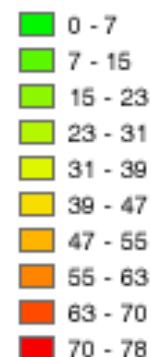
373

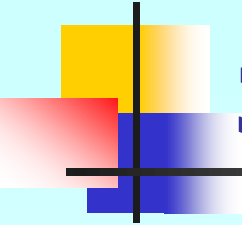


输入高程栅格



输出坡度栅格
(以度为单位)





3.7.4 Slop Calculation

$$Slope = \tan \sqrt{Slope_{we}^2 + Slope_{sn}^2}$$

$$Aspect = Slope_{sn} / Slope_{we}$$

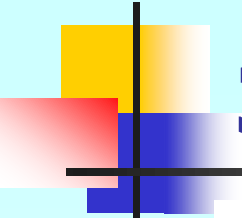
e5	e2	e6
e1	e	e3
e8	e4	e7

Slope为坡度，Aspect为坡向；

Slope_{we}为X(w-e)方向的坡度，Slope_{sn}为Y(s-n)方向的坡度

可以使用 3 x 3 像元的邻域（移动窗口）来执行计算。对于每个邻域，如果中心像元为 **NoData**，则输出也为 **NoData**。计算还要求至少 7 个与处理像元相邻的像元具有有效值。如果有效像元少于 7 个，则不执行计算，并且该处理像元处的输出将为 **NoData**。

输出栅格最外侧行列的像元为 **NoData**。这是因为沿着输入数据集边界，这些像元没有足够的有效相邻像元。



3.7.4 Slop Calculation

$$\text{a) } Slope_{we} = \frac{e_1 - e_3}{2 \times Cellsize}$$

$$\text{b) } Slope_{sn} = \frac{e_4 - e_2}{2 \times Cellsize}$$

注意单位的一致性。

$$\text{a) } Slope_{we} = \frac{(e_8 + 2e_1 + e_5) - (e_7 + 2e_3 + e_6)}{8 \times Cellsize}$$

$$\text{b) } Slope_{sn} = \frac{(e_7 + 2e_4 + e_8) - (e_6 + 2e_2 + e_5)}{8 \times Cellsize}$$

ArcMap采用的是算法2

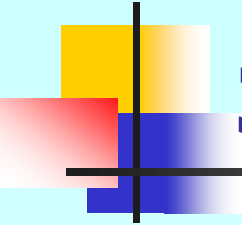
$$\text{a) } Slope_{we} = \frac{(e_8 + \sqrt{2}e_1 + e_5) - (e_7 + \sqrt{2}e_3 + e_6)}{8 \times Cellsize}$$

$$\text{b) } Slope_{sn} = \frac{(e_7 + \sqrt{2}e_4 + e_8) - (e_6 + \sqrt{2}e_2 + e_5)}{8 \times Cellsize}$$

$$\text{a) } Slope_{we} = \frac{(e_8 + e_1 + e_5) - (e_7 + e_3 + e_6)}{8 \times Cellsize}$$

$$\text{b) } Slope_{sn} = \frac{(e_7 + e_4 + e_8) - (e_6 + e_2 + e_5)}{8 \times Cellsize}$$

ERDAS Imagine
中采用的是算法4



3.7.4 Slop Calculation

面坡度算法

- 坡度：从中心像元开始在水平 (dz/dx)和垂直 (dz/dy) 方向上的变化率
- $slope_radians = ATAN (\sqrt{([dz/dx]^2 + [dz/dy]^2)})$
- $slope_degrees = ATAN (\sqrt{([dz/dx]^2 + [dz/dy]^2)}) * 57.29578$; $57.29578 = 180/\pi$

a	b	c
d	e	f
g	h	i

像元 e 在 x 方向上的变化率将通过以下算法进行计算：

- $[dz/dx] = ((c + 2f + i)*4/wght1 - (a + 2d + g)*4/wght2) / (8 * x_cellsize)$
- 其中： wght1 和 wght2 是有效像元的水平加权计数。如果：
 - c、f 和 i 均有一个有效值，则 $wght1 = (1+2*1+1) = 4$ 。
 - i 为 NoData，则 $wght1 = (1+2*1+0) = 3$ 。
 - f 为 NoData，则 $wght1 = (1+2*0+1) = 2$ 。
- 除邻域位置为 a、d 和 g 以外，类似的逻辑也适用于 wght2。
- 像元 e 在 y 方向上的变化率使用以下算法进行计算：
 - $[dz/dy] = ((g + 2h + i)*4/wght3 - (a + 2b + c)*4/wght4) / (8 * y_cellsize)$
 - 其中： wght3 和 wght4 与 $[dz/dx]$ 计算中的概念相同。

3.7.4 Slop Calculation

50	45	50
30	30	30
8	10	10

$$[dz/dx] = ((c + 2f + i)*4/wght1 - (a + 2d + g)*4/wght2) / (8 * x_cellsize) = ((50 + 60 + 10)*4/(1+2+1) - (50 + 60 + 8)*4/(1+2+1)) / (8 * 5) = (120 - 118) / 40 = 0.05$$

$$[dz/dy] = ((g + 2h + i)*4/wght3 - (a + 2b + c)*4/wght4) / (8 * y_cellsize) = ((8 + 20 + 10)*4/(1+2+1) - (50 + 90 + 50)*4/(1+2+1)) / (8 * 5) = (38 - 190) / 40 = -3.8$$

代入 x 方向和 y 方向上的变化率，使用下列算法来计算中心像元 *e* 的坡度：

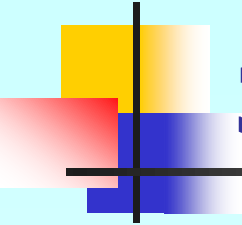
$$rise_run = \sqrt{[dz/dx]^2 + [dz/dy]^2} = \sqrt{((0.05)^2 + (-3.8)^2)} = \sqrt{(0.0025 + 14.44)} =$$

$$3.80032 \text{ slope_degrees} = \text{ATAN}(rise_run) * 57.29578 = \text{ATAN}(3.80032) * 57.29578 = 1.31349 * 57.29578 = 75.25762$$

$$57.29578 = 1.31349 * 57.29578 = 75.25762$$

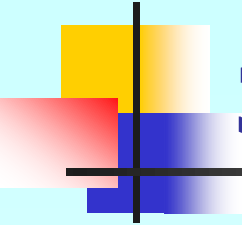
像元*e* 的整型坡度值为 75 度。

59	56	59
71	75	70
60	63	57



3.7.4 Slop Calculation

6	6	4	7	9
7	9	5	11	8
8	8	12	10	7
9	10	12	14	14



3.7.5 Application

- 依据国际地理学联合会地貌调查与地貌制图委员会关于地貌详图应用的坡地分类来划分坡度等级，规定： $0^{\circ} \sim 0.5^{\circ}$ 为平原， $>0.5^{\circ} \sim 2^{\circ}$ 为微斜坡， $>2^{\circ} \sim 5^{\circ}$ 为缓斜坡， $>5^{\circ} \sim 15^{\circ}$ 为斜坡， $>15^{\circ} \sim 35^{\circ}$ 为陡坡， $>35^{\circ} \sim 55^{\circ}$ 为峭坡， $>55^{\circ} \sim 90^{\circ}$ 为垂直壁。
- 我国耕地坡度分为 $\leq 2^{\circ}$ 、 $2^{\circ} \sim 6^{\circ}$ 、 $6^{\circ} \sim 15^{\circ}$ 、 $15^{\circ} \sim 25^{\circ}$ 、 $>25^{\circ}$ 五个坡度级（上含下不含）；规定 $>25^{\circ}$ 不能耕种。
- 西北黄土高原地区 15° 和 25° 分别为坡面流水面状侵蚀的下限和上限临界坡角。

3.7.5 Application--Cut/Fill calculation

Cut/Fill [?] [X]

Before surface: denggx_10.img [Folder icon]

After surface: slope1 [Folder icon]

Z factor: 1

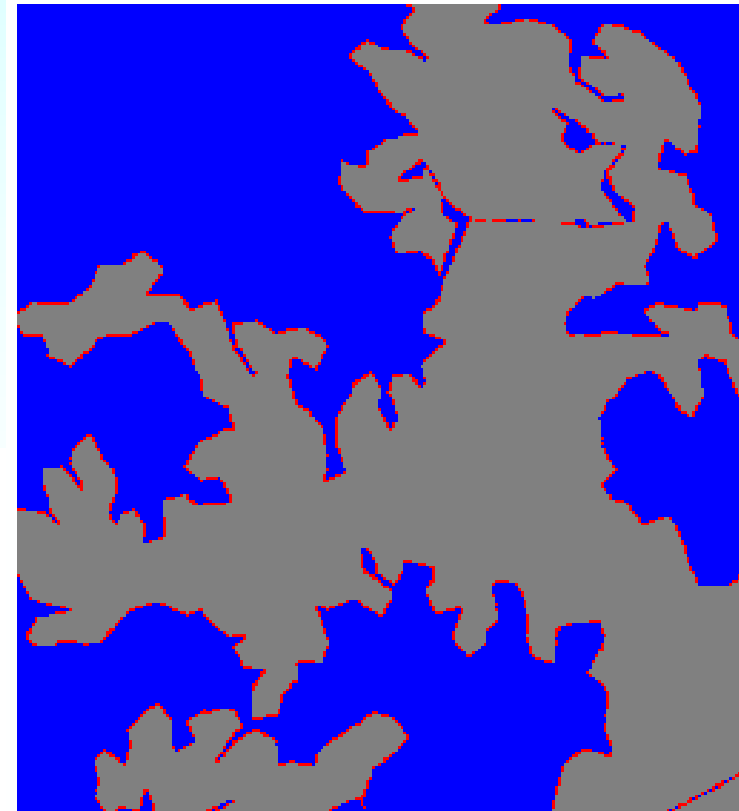
Output cell size: 30

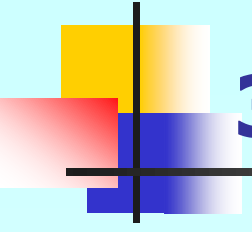
Output raster: <Temporary> [Folder icon]

[OK] [Cancel]

Cutfills between
VOLUME

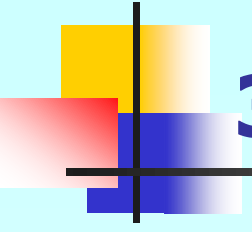
- [Red] Net Gain
- [Gray] Unchanged
- [Blue] Net Loss





3.8 Statistical analysis

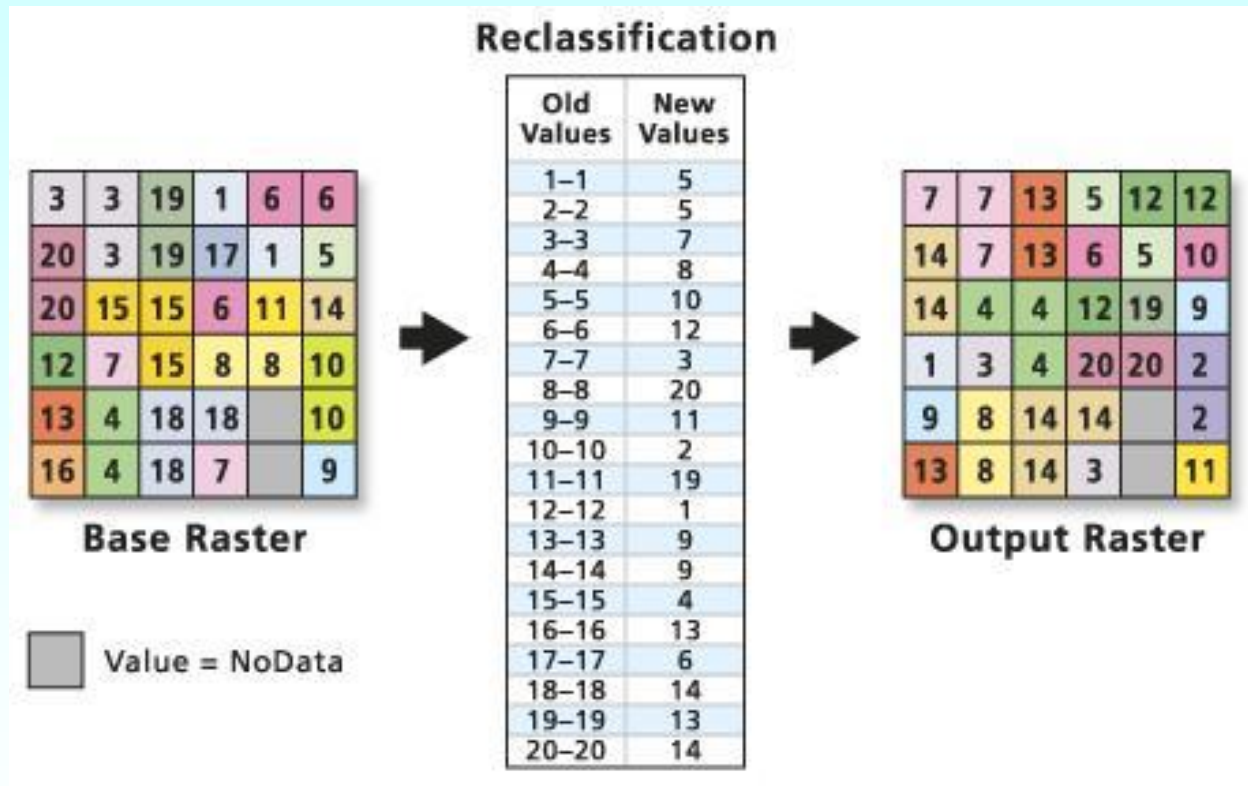
- Cell Statistics
- Neighboring Statistics
- Zonal Statistics



3.9 Reclassification

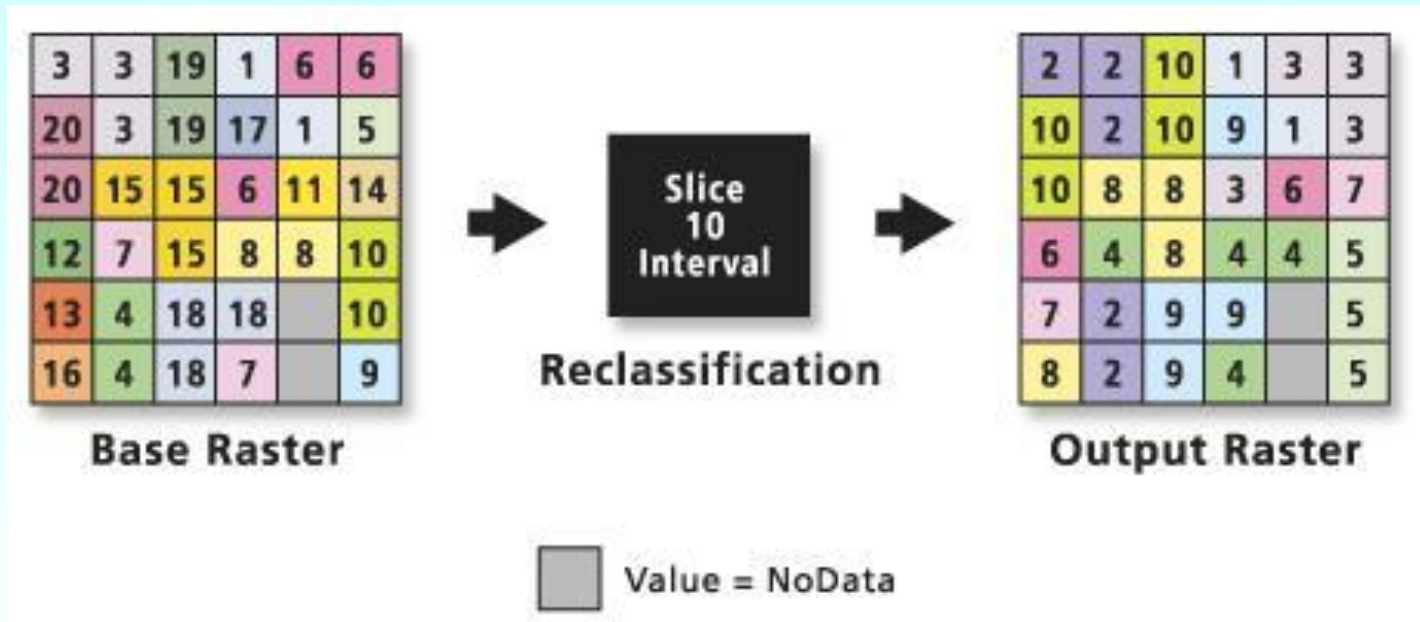
- replaced with new value
- old value group
- reclassify
- NoData setting

3.9 Reclassification



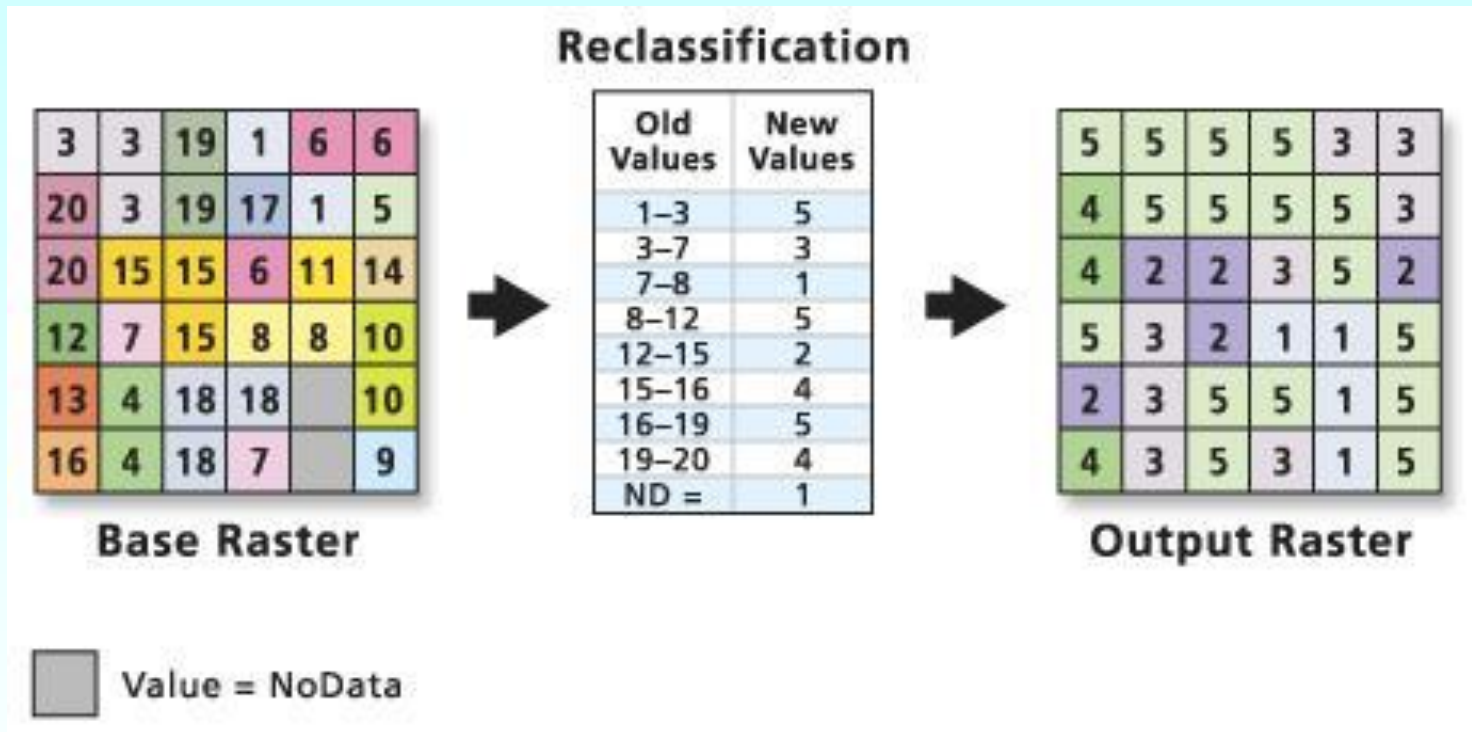
Reclass by individual values

3.9 Reclassification

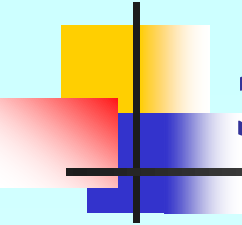


Grouping values into intervals or by area

3.9 Reclassification

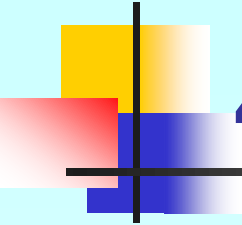


Reclass by ranges of values



3.9 Raster calculation

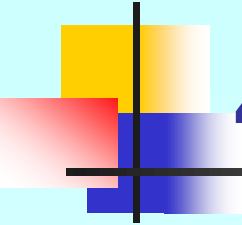
- Mathematical calculation
- function calculation
- Raster Calculator



4. Integrated Analysis: Map Algebra

Map Algebra is the analysis language for Spatial Analyst. It is a simple syntax that is similar to any algebra.

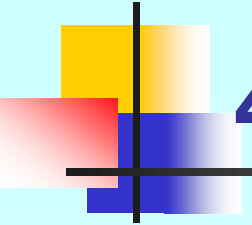
An output raster dataset will result from some manipulation of the input. The input can be as simple as a single raster dataset, raster layer, feature dataset, feature layer, or shapefile, and the manipulation can be calculating the sine of each of the location's values, or there can be a series of input raster datasets or raster layers that the manipulation is applied to such as when adding three raster datasets or raster layers together. The Map Algebra also allows you to build complex expressions and process them as a single command.



4. Integrated Analysis: Map Algebra

There are four main ways to access Map Algebra.

1. In Geoprocessing Map Algebra is accessed through
 1. the Single Output Map Algebra;
 2. and Multi Output Map Algebra tools.
2. In the Spatial Analyst toolbar it is accessed through the Raster Calculator
3. In workstation ArcGrid through the command line prompt.



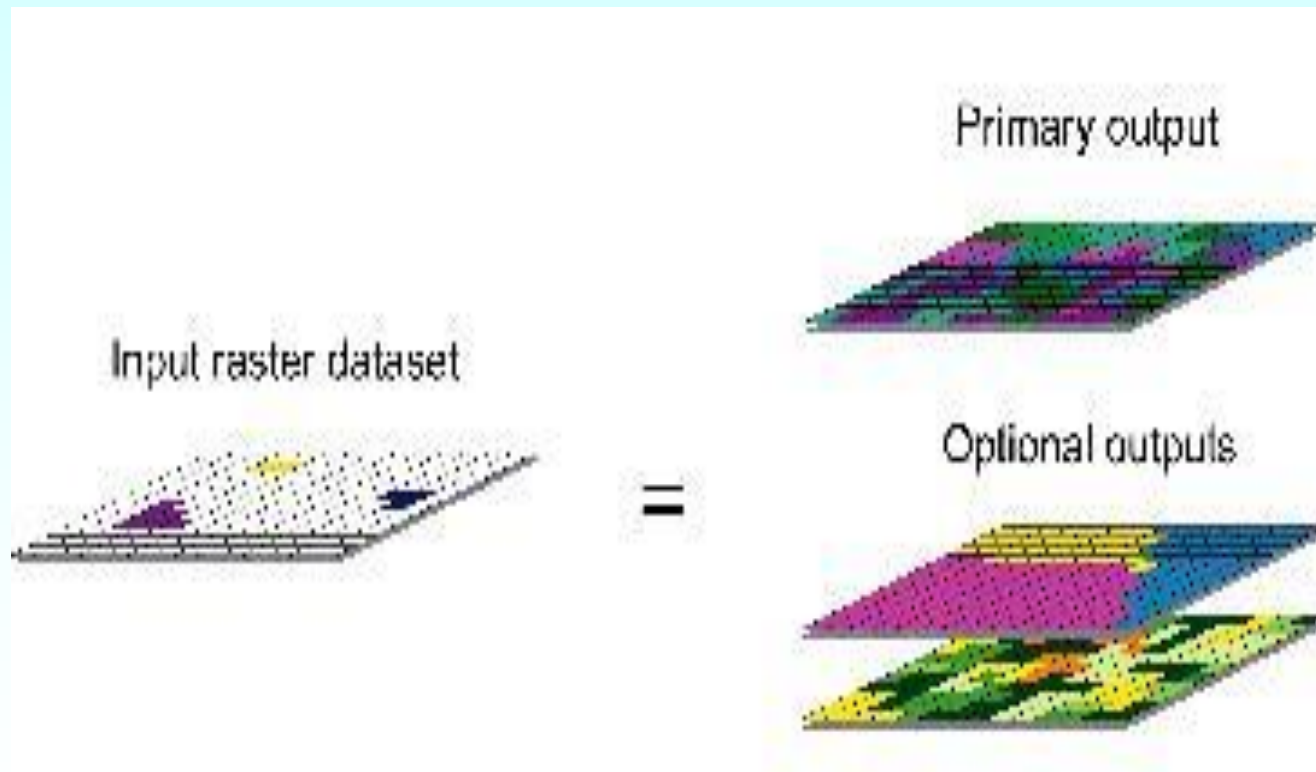
4. Integrated Analysis: Map Algebra

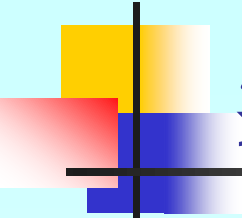
Examples:

- $[inlayer1] * [inlayer2] \text{ div } c:\backslash data \backslash inraster3$
- $[inlayer1] \& [inlayer2]$
- $[inlayer1] + c:\backslash results \backslash inraster2 - [inlayer3]$
- $([inlayer1] \text{ div } [inlayer2]) * [inlayer3]$
- $[inlayer1] + ([inlayer2] + 8)$

4. Integrated Analysis: Map Algebra

Examples:



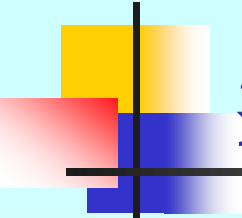


实验报告四

一、准备数据

根据实验要求，自己准备好实验数据或用课程QQ群里面共享的实验数据。注意点：

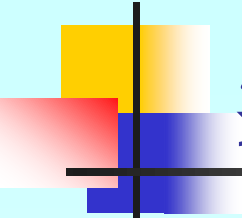
1. 明确的空间参考；
2. 所有实验数据及过程和结果都放在一个目录中，建议定义一个Dataset中；
3. 文件名命名简洁规范，便于操作结果的保持、辨识；



实验报告四

二、栅格数据基本操作实验

1. 数据：截取 $5 \times 5 \sim 10 \times 10$ 个像元的栅格数据图层2~3个图层，属性值为整数个位，像元大小为容易计算的整数，如5m，10m，保存为TIFF或IMG格式（建议不要用GRID格式）；注意正确的空间参考。
2. 通过实验，重点考察和理解PPT中讲到的基本操作：原理、计算过程和注意点
 - 认识栅格数据基本属性；
 - 基于栅格数据的距离量算；
 - 基本的数学运算、逻辑运算、位运算、其他函数运算；
 - 栅格数据的水流方向计算；
 - 坡度、坡向计算及其边缘效应
 - 最小成本路径计算
3. 举例说明栅格计算有关函数的应用场景、应用意义。



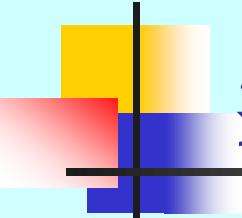
实验报告四

以下三、四两部分，选其中一个写实验报告。

三、栅格数据的最短路径（源汇分析）分析

用DEM原始数据，裁切其中一小块区域（如 20×20 个像元），做The Shortest Route分析。分别做以下三种情况：

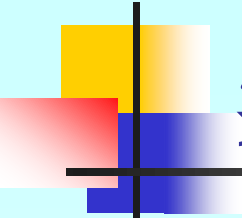
- （1）For Each Cell: 为源中每一个单元点寻找一条成本最小路径。
- （2）For Each Zone: 为每个源寻找一条成本最小路径，源中所有单元共享同一条路径。
- （3）For Each Zone Best Single: 为所有源寻找一条成本最小路径，此时，只有一个源与一个相应的目标点或目标组相连。要求：
 - 针对上述三种情况仔细分析这种分析方法的原理；注意要用中间步骤的图、或局部放大的图来说明。
 - 说明分析的技术过程，步骤，涉及到的相关参数的对结果的影响。
 - 举例说明上述三种情况的用途。



实验报告四

四、基于栅格数据的地图代数分析方法

- (1) 总结地图代数的计算公式，举例说明其他计算方法及应用意义；
- (2) 用给定的大学城**DEM**数据（或自己有的其他区域、软件工程课程设计的案例区域），选择其中一个区块（如**2×3**平方千米），下载**2-3**个时相的遥感影像，解译或不解译均可，根据前述应用意义，自己设定一些应用问题（如加油站选址规划、校园内部空间格局动态变化分析等等），利用地图代数方法给出答案。要给出详细的中间过程。



实验报告四

五、总结并撰写实验报告

根据选做的实验，实验报告应包含（二）+（三）或（二）+（四）内容。

1. 回答PPT中的思考题和考试练习题；
2. 碰到的技术细节问题以及解决办法；
3. 分析技术方法的应用；
4. 延伸的有关思考。